Past Exam & Quiz Questions — Midterm II Compilation (MATH 205 Linear Algebra)

This document compiles **every single relevant question and proof** from all files in `past assessments/` (12 exam/solution files and 13 unique quiz solution files) covering the **Midterm II syllabus**:

1. **Vectors in $\mathbb{R}^n$, Dot Product, Norm, Distance, Orthogonality & Inner Product Inequalities**
2. **General Vector Spaces (10 Axioms) & Arithmetic Properties**
3. **Subspaces & The Subspace Test**
4. **Linear Combinations, Span, Linear Independence/Dependence, Bases & Dimension**
5. **Fundamental Matrix Subspaces ($\mathrm{Row}, \mathrm{Col}, \mathrm{Null}$), Rank–Nullity & Consistency Bridges**
6. **Linear Transformations, Standard Matrices, Kernel ($\ker T$), Range ($R(T)$), One-to-One & Onto**

> **Coverage Note:** All 29 files (25 unique after removing exact MD5 duplicates) in `past assessments/` were scanned, including embedded raster/vector equations and `.docx` solution images. Official grading-key solutions are integrated and verified for arithmetic accuracy; printed typos in old exam papers are flagged explicitly. Pure Chapter 1–2 matrix/determinant/LU/Cramer questions (pre-Midterm I) and Chapter 6–7 Gram–Schmidt / Orthogonal Matrix / Eigenvalue / Diagonalization questions (Final-only topics) are listed briefly in the Appendix at the end.

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High-Yield `[PROOF]` Checklist (Every Proof Seen in Past Exams/Quizzes)

ID	Theorem / Claim	Source(s)
Q1.1	$(\mathbf{u}+\mathbf{v})\cdot\mathbf{w} = \mathbf{u}\cdot\mathbf{w} + \mathbf{v}\cdot\mathbf{w}$	Sp26 Mid B Q3(a)
Q1.2	$A\mathbf{u}\cdot\mathbf{v} = \mathbf{u}\cdot A^T\mathbf{v}$ and $\mathbf{u}\cdot A\mathbf{v} = A^T\mathbf{u}\cdot\mathbf{v}$	F22 Mid Q9(a), Sp26 `MID+B` Q5
Q1.3	Unit vector $\frac{1}{\lvert\mathbf{v}\rvert}\mathbf{v}$ has norm 1	Sp24 Mid B Q2(a)
Q1.4	Polarization & Parallelogram identities	F22 FE Q1(a), Sp23 FE Q3(b), Quiz 11 P1
Q1.6	General Cauchy–Schwarz: $\lvert\langle\mathbf{u},\mathbf{v}\rangle\rvert \le \lvert\mathbf{u}\rvert\lvert\mathbf{v}\rvert$	Sp22 FE Q7(a), Sp23 FE Q3(a), Sp24 FE Q8(b), Quiz 11 P2(a)
Q1.7	Integral Cauchy–Schwarz on $C[0,1]$	F22 FE Q7
Q1.8	Matrix Cauchy–Schwarz: $(\mathbf{v}^TA^TA\mathbf{u})^2 \le (\mathbf{u}^TA^TA\mathbf{u})(\mathbf{v}^TA^TA\mathbf{v})$	Sp24 Mid B Q5, Sp25 FE P4(a)
Q1.9	Triangle inequality $\lvert\mathbf{u}+\mathbf{v}\rvert \le \lvert\mathbf{u}\rvert + \lvert\mathbf{v}\rvert$ & norm properties	Sp22 Mid Q5(a), F22 Mid Q9(b), Sp26 Mid B Q3(b), Sp22 FE Q7(b), Quiz 11 P5
Q1.10	Distance triangle inequality $d(\mathbf{u},\mathbf{v}) \le d(\mathbf{u},\mathbf{w}) + d(\mathbf{w},\mathbf{v})$	Sp22 Mid Q5(b), F22 FE Q1(b)
Q1.11	Pythagorean theorem for 2 and r pairwise orthogonal vectors	Sp24 Mid B Q2(b), Quiz 11 P2(b)
Q1.12	$\mathbf{w}\perp\mathbf{u}_1,\mathbf{u}_2 \implies \mathbf{w}\perp k_1\mathbf{u}_1+k_2\mathbf{u}_2$; $A\mathbf{x}=\mathbf{0} \iff \mathbf{x}\perp$ rows	Sp22 FE Q10, Quiz 11 P6
Q2.1	Zero vector space $V=\{\mathbf{0}\}$ satisfies all 10 axioms	Sp24 Mid B Q3
Q2.2	Positive reals $(0,\infty)$ under $u+v=uv$, $ku=u^k$ (all 10 axioms / Ax 4–5)	Sp23 Mid Q5, Sp22 Mid Q6(b), F22 FE Q2(a), Sp24 FE Q3

ID	Theorem / Claim	Source(s)
Q2.4	Can a vector space have only 2 distinct elements? (Disproof)	Sp22 Mid Q6(a)
Q2.5	$k\mathbf{0} = \mathbf{0}$ and $(-1)\mathbf{u} = -\mathbf{u}$ in any vector space V	F22 FE Q3(a),(b)
Q3.1	Two-condition Subspace Test ($W \neq \emptyset$, closed under $+$, scalar mult)	F22 Mid Q7, Sp24 FE Q5
Q3.2	One-step subspace test ($k\mathbf{u} + \mathbf{v} \in W$); $W_1 \cap W_2$ is a subspace	F22 Mid Q8(a)(i), Sp22 FE Q4(a),(e)
Q3.3–3.5	Subspace verifications (P_n even, $p'(0) = 0$, $p'(1) = 0$, matrices, integrals)	Sp25 FE P9(a),(b), Sp22 Mid Q10, Quiz 6 P1, P4–P7
Q4.1	$\text{span}\{v_1, \ldots, v_r\}$ is the smallest subspace containing $v_1, \ldots, v_r$	F22 FE Q8(a),(b)
Q4.5	Two vectors LI $\iff$ neither is a scalar multiple; $\{\mathbf{a}-\mathbf{b}, \mathbf{b}-\mathbf{c}, \mathbf{c}-\mathbf{a}\}$ LD	F22 FE Q2(b), Sp22 Mid Q7(a)
Q4.6	S LI and $\mathbf{c} \notin \text{span}(S) \implies S \cup \{\mathbf{c}\}$ LI; S LI $\iff$ no vector is LC of others	Sp22 Mid Q7(b), Q8
Q4.7	Orthogonal set of nonzero vectors is linearly independent	Sp23 FE Q8, Quiz 11 P4(a)
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Q4.10	Half-Test ($\lvert S\rvert = n$); $\dim(W) \leq \dim(V)$ and $\dim(W) = \dim(V) \implies W = V$	Sp22 FE Q5(a),(b), Sp24 FE Q4
Q5.1	Solution structure of $A\mathbf{x} = \mathbf{b}$ as $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_0$	Sp22 Mid Q4
Q5.2	$A\mathbf{x} = \mathbf{b}$ consistent $\iff \mathbf{b} \in \text{Col}(A)$	Sp23 FE Q2(a), Sp25 FE P4(b)
Q5.3	$A\mathbf{x} = \mathbf{b}$ consistent $\forall \mathbf{b} \iff A$ invertible; Master Theorem 4-way cycle	Sp22 FE Q8, Sp24 Mid A Q2, Sp24 FE Q1, Sp23 Mid Q7
Q5.4	$\dim(\text{Row}(A)) = \dim(\text{Col}(A))$ and $\text{rank}(A) = \text{rank}(A^T)$	Sp23 FE Q1(a),(b)
Q6.1	$T(A) = \det(A)$ nonlinear; T 1-to-1 $\iff \ker(T) = \{\mathbf{0}\}$; matrix rep; $\dim R(T_A) = \text{rank}(A)$	Sp23 FE Q6(c)–(e), Q2(b)

Part 1: Vectors, Inner Products, Norms & Inequalities

Q1.1 [PROOF] Linearity of Euclidean Dot Product

Source: *Spring 2026 Midterm Part B Q3(a) [4 Marks] (+ Official Sol in `MID+B+Solution.docx`)*

> If $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are vectors in $\mathbb{R}^n$, prove that:
>
> $$(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$$

Solution: Let $\mathbf{u} = (u_1, u_2, \ldots, u_n)$, $\mathbf{v} = (v_1, v_2, \ldots, v_n)$, and $\mathbf{w} = (w_1, w_2, \ldots, w_n)$. Then:

$$\begin{aligned}(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} &= (u_1 + v_1,\ u_2 + v_2,\ \ldots,\ u_n + v_n) \cdot (w_1, w_2, \ldots, w_n) \\ &= (u_1 + v_1)w_1 + (u_2 + v_2)w_2 + \cdots + (u_n + v_n)w_n \\ &= (u_1w_1 + u_2w_2 + \cdots + u_nw_n) + (v_1w_1 + v_2w_2 + \cdots + v_nw_n) \\ &= \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}. \quad \blacksquare\end{aligned}$$

Q1.2 [PROOF] Matrix Adjoint Identity for Euclidean Dot Product

Source: *Fall 2022 Midterm Q9(a) [~5 Marks];* `MID+B+Solution.docx` *Q5*

> Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in $\mathbb{R}^n$ (treated as $n \times 1$ column matrices) and let A be an $n \times n$ matrix. Show:
>
> - **(Fall 22 Mid Q9a):** $A\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot A^T\mathbf{v}$

- **(Sp26 MID+B Q5):** $\mathbf{u} \cdot A\mathbf{v} = A^T\mathbf{u} \cdot \mathbf{v}$

Solution: Using the matrix formula for the Euclidean dot product $\mathbf{a} \cdot \mathbf{b} = \mathbf{b}^T\mathbf{a}$ (or equivalently $\mathbf{a}^T\mathbf{b}$):

For $A\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot A^T\mathbf{v}$:

$$A\mathbf{u} \cdot \mathbf{v} = (A\mathbf{u})^T\mathbf{v} = (\mathbf{u}^T A^T)\mathbf{v} = \mathbf{u}^T(A^T\mathbf{v}) = \mathbf{u} \cdot (A^T\mathbf{v}). \quad \blacksquare$$

For $\mathbf{u} \cdot A\mathbf{v} = A^T\mathbf{u} \cdot \mathbf{v}$ (official MID+B proof using $\mathbf{a} \cdot \mathbf{b} = \mathbf{b}^T\mathbf{a}$):

$$\mathbf{u} \cdot A\mathbf{v} = (A\mathbf{v})^T\mathbf{u} = \mathbf{v}^T(A^T\mathbf{u}) = (A^T\mathbf{u}) \cdot (\mathbf{v}^T)^T = A^T\mathbf{u} \cdot \mathbf{v}. \quad \blacksquare$$

(Properties used: $(AB)^T = B^T A^T$, associativity of matrix multiplication, and $(\mathbf{v}^T)^T = \mathbf{v}$.)

Q1.3 [PROOF] Normalization of a Nonzero Vector

Source: *Spring 2024 Midterm Part B Q2(a) [4 Marks] (+ Official Sol)*

> Show that if $\mathbf{v}$ is a nonzero vector in $\mathbb{R}^n$, then $\dfrac{1}{\|\mathbf{v}\|}\mathbf{v}$ has Euclidean norm 1.

Solution: Since $\mathbf{v} = (v_1, v_2, \ldots, v_n) \neq \mathbf{0}$, we have $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} > 0$, so:

$$\frac{1}{\|\mathbf{v}\|}\mathbf{v} = \left(\frac{v_1}{\|\mathbf{v}\|}, \frac{v_2}{\|\mathbf{v}\|}, \ldots, \frac{v_n}{\|\mathbf{v}\|}\right).$$

Taking its Euclidean norm:

$$\left\|\frac{1}{\|\mathbf{v}\|}\mathbf{v}\right\| = \sqrt{\left(\frac{v_1}{\|\mathbf{v}\|}\right)^2 + \left(\frac{v_2}{\|\mathbf{v}\|}\right)^2 + \cdots + \left(\frac{v_n}{\|\mathbf{v}\|}\right)^2} = \sqrt{\frac{v_1^2 + v_2^2 + \cdots + v_n^2}{v_1^2 + v_2^2 + \cdots + v_n^2}} = \sqrt{1} = 1. \quad \blacksquare$$

(Shorter inner-product version: $\left\|\frac{1}{\|\mathbf{v}\|}\mathbf{v}\right\| = \left|\frac{1}{\|\mathbf{v}\|}\right| \|\mathbf{v}\| = \frac{\|\mathbf{v}\|}{\|\mathbf{v}\|} = 1$.)

Q1.4 [PROOF] Polarization Identity & Parallelogram Law

Sources:

- *Fall 2022 FE Q1(a) [5 Marks]* ($\mathbf{u}, \mathbf{v} \in \mathbb{R}^k$: $\mathbf{u} \cdot \mathbf{v} = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2)$)
- *Spring 2023 FE Q3(b) [5 Marks]* (General inner product space V: $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\|\mathbf{u} + \mathbf{v}\|^2 - \frac{1}{4}\|\mathbf{u} - \mathbf{v}\|^2$)
- *Quiz 11 Problem 1(a),(b) (+ Official Sol)* ($\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$; note Quiz 11 printed missing squares on the RHS, proved with squares in the official key)

> Let V be an inner product space (or $\mathbb{R}^k$ with Euclidean dot product) and let $\mathbf{u}, \mathbf{v} \in V$. Prove:
>
> - **(a) Parallelogram Law:** $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$
> - **(b) Polarization Identity:** $\langle \mathbf{u}, \mathbf{v} \rangle = \dfrac{1}{4}\left(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2\right)$

Solution: Using the norm definition $\|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{w} \rangle$ along with symmetry and bilinearity (additivity + homogeneity) of the inner product:

$$\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2,$$
$$\|\mathbf{u} - \mathbf{v}\|^2 = \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle - \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{u}\|^2 - 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2.$$

- **(a) Add the two equations:**

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2. \quad \blacksquare$$

- **(b) Subtract the second from the first:**

$$\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 = 4\langle \mathbf{u}, \mathbf{v} \rangle \implies \langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\left(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2\right). \quad \blacksquare$$

(For $\mathbb{R}^k$ specifically, write $\mathbf{u} \cdot \mathbf{v}$ in place of $\langle \mathbf{u}, \mathbf{v} \rangle$.)

Q1.5 [CALC] Computing Dot Product from Norms

Source: *Quiz 6 Problem 2 (+ Official Sol)*

> Find $\mathbf{u} \cdot \mathbf{v}$ given that $\|\mathbf{u}+\mathbf{v}\| = 3$ and $\|\mathbf{u}-\mathbf{v}\| = 1$.

Solution: Using $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2(\mathbf{u}\cdot\mathbf{v}) + \|\mathbf{v}\|^2 = 9$ and $\|\mathbf{u}-\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2(\mathbf{u}\cdot\mathbf{v}) + \|\mathbf{v}\|^2 = 1$, subtract the two equations (Polarization Identity):

$$4(\mathbf{u}\cdot\mathbf{v}) = 9 - 1 = 8 \implies \mathbf{u}\cdot\mathbf{v} = 2.$$

Q1.6 [PROOF] Cauchy–Schwarz Inequality (General Inner Product Space)

Sources: *Spring 2022 FE Q7(a) [7 Marks]; Spring 2023 FE Q3(a) [5 Marks]; Spring 2024 FE Q8(b) [6 Marks]; Quiz 11 P2(a)*

> Let V be a real inner product space with $\mathbf{u}, \mathbf{v} \in V$. Prove that:
>
> $$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\,\|\mathbf{v}\|$$
>
> *(Note: Spring 2022 FE Q7(a) had a printed typo writing $\|\mathbf{x}\| = \langle \mathbf{x}, \mathbf{x}\rangle$ instead of $\sqrt{\langle \mathbf{x}, \mathbf{x}\rangle}$. In standard exams use $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x}\rangle}$.)*

Solution: Case 1: $\mathbf{u} = \mathbf{0}$. Then $\|\mathbf{u}\| = 0$ and $\langle \mathbf{0}, \mathbf{v}\rangle = 0$ (since $\langle 0\mathbf{u}, \mathbf{v}\rangle = 0\langle \mathbf{u}, \mathbf{v}\rangle = 0$). Both sides equal 0, so $0 \leq 0$ holds.

Case 2: $\mathbf{u} \neq \mathbf{0}$. Then $\langle \mathbf{u}, \mathbf{u}\rangle = \|\mathbf{u}\|^2 > 0$. Let $t \in \mathbb{R}$ be any real scalar, and consider $\mathbf{w} = t\mathbf{u} + \mathbf{v}$. By positivity of the inner product:

$$\langle t\mathbf{u} + \mathbf{v},\ t\mathbf{u} + \mathbf{v}\rangle \geq 0.$$

Expand using bilinearity and symmetry:

$$t^2\langle \mathbf{u}, \mathbf{u}\rangle + 2t\langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{v}, \mathbf{v}\rangle \geq 0 \implies at^2 + bt + c \geq 0$$

where $a = \langle \mathbf{u}, \mathbf{u}\rangle = \|\mathbf{u}\|^2 > 0$, $b = 2\langle \mathbf{u}, \mathbf{v}\rangle$, and $c = \langle \mathbf{v}, \mathbf{v}\rangle = \|\mathbf{v}\|^2$. Since this quadratic in t is always ≥ 0 and opens upward ($a > 0$), it can have at most one real root, so its discriminant must satisfy $\Delta \leq 0$:

$$b^2 - 4ac \leq 0 \implies (2\langle \mathbf{u}, \mathbf{v}\rangle)^2 - 4\|\mathbf{u}\|^2\|\mathbf{v}\|^2 \leq 0 \implies \langle \mathbf{u}, \mathbf{v}\rangle^2 \leq \|\mathbf{u}\|^2\|\mathbf{v}\|^2.$$

Taking square roots of both nonnegative sides gives:

$$|\langle \mathbf{u}, \mathbf{v}\rangle| \leq \|\mathbf{u}\|\,\|\mathbf{v}\|. \quad \blacksquare$$

(Equality holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.)

Q1.7 [PROOF] Integral Cauchy–Schwarz Inequality on $C[0,1]$

Source: *Fall 2022 FE Q7 [10 Marks]*

> Let $f(x)$ and $g(x)$ be continuous functions on $[0,1]$. Prove that:
>
> $$\left(\int_0^1 f(x)g(x)\,dx\right)^2 \leq \left(\int_0^1 f^2(x)\,dx\right)\left(\int_0^1 g^2(x)\,dx\right)$$

Solution: Let $V = C[0,1]$ be equipped with the inner product $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$, or prove directly:

- **If $\int_0^1 f^2(x)\,dx = 0$:** Since $f^2(x) \geq 0$ is continuous on $[0,1]$, this forces $f(x) \equiv 0$ on $[0,1]$, so both sides equal 0 and the inequality holds.
- **If $\int_0^1 f^2(x)\,dx > 0$:** For every $t \in \mathbb{R}$, $(tf(x) + g(x))^2 \geq 0$ for all $x \in [0,1]$, hence:

  $$\int_0^1 (tf(x) + g(x))^2\,dx \geq 0.$$

  Expanding the integrand and using linearity of the definite integral:

  $$t^2\left(\int_0^1 f^2(x)\,dx\right) + 2t\left(\int_0^1 f(x)g(x)\,dx\right) + \left(\int_0^1 g^2(x)\,dx\right) \geq 0 \quad \forall t \in \mathbb{R}.$$

  This quadratic $at^2 + bt + c \geq 0$ (with $a = \int_0^1 f^2(x)\,dx > 0$) has non-positive discriminant $b^2 - 4ac \leq 0$:

$$4\left(\int_0^1 f(x)g(x)\,dx\right)^2 - 4\left(\int_0^1 f^2(x)\,dx\right)\left(\int_0^1 g^2(x)\,dx\right) \le 0,$$

so:

$$\left(\int_0^1 f(x)g(x)\,dx\right)^2 \le \left(\int_0^1 f^2(x)\,dx\right)\left(\int_0^1 g^2(x)\,dx\right). \quad \blacksquare$$

Q1.8 [PROOF] Matrix Cauchy–Schwarz Inequality

Sources: *Spring 2024 Midterm Part B Q5 [10 Marks] (+ Official Sol); Spring 2025 FE Solution Problem 4a*

> Prove: If $\mathbf{u}$ and $\mathbf{v}$ are $n \times 1$ matrices and A is an $n \times n$ matrix, then:
>
> $$(\mathbf{v}^T A^T A\mathbf{u})^2 \le (\mathbf{u}^T A^T A\mathbf{u})(\mathbf{v}^T A^T A\mathbf{v})$$
>
> *(Typo alert: Spring 2024 Midterm Part B printed `(\mathbf{u}^T A^T A \mathbf{v})` at the far right instead of `(\mathbf{v}^T A^T A \mathbf{v})`; the Spring 2025 FE Solution corrected it to $\mathbf{v}^T A^T A\mathbf{v}$.)*

Solution (Official Spring 2025 FE Key): Let $\mathbf{x} = A\mathbf{u}$ and $\mathbf{y} = A\mathbf{v}$ in $\mathbb{R}^n$. Observe that:

$$\mathbf{y} \cdot \mathbf{x} = \mathbf{y}^T\mathbf{x} = (A\mathbf{v})^T(A\mathbf{u}) = \mathbf{v}^T A^T A\mathbf{u}.$$

Similarly,

$$\|\mathbf{x}\|^2 = \mathbf{x}^T\mathbf{x} = (A\mathbf{u})^T(A\mathbf{u}) = \mathbf{u}^T A^T A\mathbf{u}, \qquad \|\mathbf{y}\|^2 = \mathbf{y}^T\mathbf{y} = (A\mathbf{v})^T(A\mathbf{v}) = \mathbf{v}^T A^T A\mathbf{v}.$$

Applying the squared Cauchy–Schwarz inequality $(\mathbf{x} \cdot \mathbf{y})^2 \le \|\mathbf{x}\|^2\|\mathbf{y}\|^2$ in $\mathbb{R}^n$:

$$(\mathbf{v}^T A^T A\mathbf{u})^2 \le (\mathbf{u}^T A^T A\mathbf{u})(\mathbf{v}^T A^T A\mathbf{v}). \quad \blacksquare$$

Q1.9 [PROOF] Triangle Inequality & Inner Product Norm Properties

Sources:

- *Spring 2022 Mid Q5(a) [5 Marks]; Fall 2022 Mid Q9(b); Spring 2026 Mid Part B Q3(b) [6 Marks] (+ `MID+B`)* (in $\mathbb{R}^n$)
- *Spring 2022 FE Q7(b) [3 Marks]; Quiz 11 Problem 5(a)–(c)* (in any inner product space V)

> Let V be an inner product space (or $\mathbb{R}^n$) and $\mathbf{u}, \mathbf{v} \in V$. Prove:
>
> - **(a)** $\|\mathbf{u}\| \ge 0$ *(Quiz 11 P5a)*
> - **(b)** $\|k\mathbf{u}\| = |k|\|\mathbf{u}\|$ *(Quiz 11 P5c)*
> - **(c) Triangle Inequality:** $\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$

Solution:

- **(a)** By definition $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u}\rangle}$, and positivity gives $\langle \mathbf{u}, \mathbf{u}\rangle \ge 0$, so $\|\mathbf{u}\| \ge 0$.
- **(b)** $\|k\mathbf{u}\| = \sqrt{\langle k\mathbf{u}, k\mathbf{u}\rangle} = \sqrt{k^2\langle \mathbf{u}, \mathbf{u}\rangle} = |k|\sqrt{\langle \mathbf{u}, \mathbf{u}\rangle} = |k|\|\mathbf{u}\|$.
- **(c) Triangle Inequality:** Compute $\|\mathbf{u} + \mathbf{v}\|^2$:

$$\begin{aligned}
\|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v}\rangle \\
&= \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v}\rangle + \|\mathbf{v}\|^2 \\
&\le \|\mathbf{u}\|^2 + 2|\langle \mathbf{u}, \mathbf{v}\rangle| + \|\mathbf{v}\|^2 \qquad (\text{since } x \le |x| \text{ for any } x \in \mathbb{R}) \\
&\le \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 \qquad (\text{by the Cauchy–Schwarz inequality}) \\
&= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2.
\end{aligned}$$

Since both $\|\mathbf{u} + \mathbf{v}\|$ and $\|\mathbf{u}\| + \|\mathbf{v}\|$ are nonnegative, taking square roots yields:

$$\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|. \quad \blacksquare$$

Q1.10 [PROOF] Triangle Inequality for Distance

Sources: *Spring 2022 Mid Q5(b) [5 Marks]; Fall 2022 FE Q1(b) [5 Marks]*

> If $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are vectors in $\mathbb{R}^n$, show that:
>
> $$d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$$

Solution: Recall the Euclidean distance $d(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|$. Write $\mathbf{u} - \mathbf{v} = (\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})$ and apply the norm triangle inequality $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ with $\mathbf{x} = \mathbf{u} - \mathbf{w}$ and $\mathbf{y} = \mathbf{w} - \mathbf{v}$:

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \|(\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})\| \leq \|\mathbf{u} - \mathbf{w}\| + \|\mathbf{w} - \mathbf{v}\| = d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v}). \quad \blacksquare$$

Q1.11 [PROOF] Pythagorean Theorem for Two and r Pairwise Orthogonal Vectors

Sources: *Quiz 11 Problem 2(b) (+ Official Sol); Spring 2024 Mid Part B Q2(b) [6 Marks] (+ Official Sol)*

> - **(a)** If $\mathbf{u} \perp \mathbf{v}$ in an inner product space V, prove that $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.
> - **(b)** Show that if $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r$ are pairwise orthogonal vectors in $\mathbb{R}^n$, then:
>
> $$\|\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_r\|^2 = \|\mathbf{v}_1\|^2 + \|\mathbf{v}_2\|^2 + \cdots + \|\mathbf{v}_r\|^2$$

Solution:

- **(a)** Since $\mathbf{u} \perp \mathbf{v}$, $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. Therefore:

$$\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

- **(b)** Since $\mathbf{v}_1$ is orthogonal to $\mathbf{v}_2, \ldots, \mathbf{v}_r$, we have $\mathbf{v}_1 \cdot (\mathbf{v}_2 + \cdots + \mathbf{v}_r) = \mathbf{v}_1 \cdot \mathbf{v}_2 + \cdots + \mathbf{v}_1 \cdot \mathbf{v}_r = 0$. Applying part (a) repeatedly (induction / $r - 1$ steps):

$$\begin{aligned} \|\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_r\|^2 &= \|\mathbf{v}_1 + (\mathbf{v}_2 + \cdots + \mathbf{v}_r)\|^2 \\ &= \|\mathbf{v}_1\|^2 + \|\mathbf{v}_2 + \cdots + \mathbf{v}_r\|^2 \\ &= \|\mathbf{v}_1\|^2 + \|\mathbf{v}_2\|^2 + \cdots + \|\mathbf{v}_r\|^2. \quad \blacksquare \end{aligned}$$

  (Equivalently, expand $(\sum_{i=1}^r \mathbf{v}_i) \cdot (\sum_{j=1}^r \mathbf{v}_j) = \sum_{i=1}^r \|\mathbf{v}_i\|^2 + \sum_{i \neq j} \mathbf{v}_i \cdot \mathbf{v}_j = \sum_{i=1}^r \|\mathbf{v}_i\|^2$ since $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ for $i \neq j$.)

Q1.12 [PROOF] Orthogonality to Span & Row-Orthogonality of $\mathrm{Null}(A)$

Sources: *Quiz 11 Problem 6 (+ Official Sol); Spring 2022 FE Q10 [10 Marks]*

> - **(a) (Quiz 11 P6):** Let V be an inner product space. Show that if $\mathbf{w}$ is orthogonal to both $\mathbf{u}_1$ and $\mathbf{u}_2$, it is orthogonal to $k_1\mathbf{u}_1 + k_2\mathbf{u}_2$ for all scalars k_1, k_2. Interpret geometrically in $\mathbb{R}^3$.
> - **(b) (Sp22 FE Q10):** Let $A\mathbf{x} = \mathbf{0}$ be a homogeneous system where A is $m \times n$. Prove that $\mathbf{x} = [x_1, x_2, \ldots, x_n]^T$ is a solution if and only if $\mathbf{x}$ is orthogonal to every row vector of A (Euclidean inner product).

Solution:

- **(a)** Suppose $\langle \mathbf{w}, \mathbf{u}_1 \rangle = 0$ and $\langle \mathbf{w}, \mathbf{u}_2 \rangle = 0$. By bilinearity of the inner product:

$$\langle \mathbf{w}, k_1\mathbf{u}_1 + k_2\mathbf{u}_2 \rangle = k_1\langle \mathbf{w}, \mathbf{u}_1 \rangle + k_2\langle \mathbf{w}, \mathbf{u}_2 \rangle = k_1(0) + k_2(0) = 0.$$

  Geometric interpretation in $\mathbb{R}^3$: If $\mathbf{u}_1, \mathbf{u}_2$ are non-collinear, $\mathrm{span}\{\mathbf{u}_1, \mathbf{u}_2\}$ is a plane through the origin; $\mathbf{w}$ is orthogonal to every vector in the plane (i.e., $\mathbf{w}$ is a normal vector to the plane).
- **(b)** Let $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_m$ be the row vectors of A. By definition of matrix multiplication, the i-th entry of $A\mathbf{x}$ is the Euclidean dot product $\mathbf{r}_i \cdot \mathbf{x}$:

$$A\mathbf{x} = \begin{bmatrix} \mathbf{r}_1 \cdot \mathbf{x} \\ \mathbf{r}_2 \cdot \mathbf{x} \\ \vdots \\ \mathbf{r}_m \cdot \mathbf{x} \end{bmatrix}.$$

  Therefore:

$$A\mathbf{x} = \mathbf{0} \iff \mathbf{r}_i \cdot \mathbf{x} = 0 \text{ for all } i = 1, \ldots, m \iff \mathbf{x} \text{ is orthogonal to every row vector of } A. \quad \blacksquare$$

Q1.13 [CALC] Orthogonal Unit Vectors in $\mathbb{R}^4$

Source: *Quiz 6 Problem 3 (+ Official Sol)*

> Find two vectors of norm 1 that are orthogonal to $\mathbf{u} = (2, 1, -4, 0)$ and $\mathbf{v} = (-1, -1, 2, 2)$.

Solution: Let $\mathbf{w} = (w_1, w_2, w_3, w_4)$. Orthogonality requires:

1. $\mathbf{u} \cdot \mathbf{w} = 2w_1 + w_2 - 4w_3 = 0 \implies w_2 = -2w_1 + 4w_3$,
2. $\mathbf{v} \cdot \mathbf{w} = -w_1 - w_2 + 2w_3 + 2w_4 = 0$.

Substitute w_2 into (2):

$$-w_1 - (-2w_1 + 4w_3) + 2w_3 + 2w_4 = 0 \implies w_1 - 2w_3 + 2w_4 = 0 \implies w_1 = 2w_3 - 2w_4.$$

Then $w_2 = -2(2w_3 - 2w_4) + 4w_3 = 4w_4$. Setting $w_3 = s, w_4 = t$ gives $\mathbf{w} = (2s - 2t,\ 4t,\ s,\ t)$.

- Take $s = 1, t = 0$: $\mathbf{a} = (2, 0, 1, 0)$ has $\|\mathbf{a}\| = \sqrt{5}$, giving $\mathbf{w}_1 = \left(\frac{2}{\sqrt{5}},\ 0,\ \frac{1}{\sqrt{5}},\ 0\right)$.
- Take $s = 0, t = 1$: $\mathbf{b} = (-2, 4, 0, 1)$ has $\|\mathbf{b}\| = \sqrt{4 + 16 + 1} = \sqrt{21}$, giving $\mathbf{w}_2 = \left(-\frac{2}{\sqrt{21}},\ \frac{4}{\sqrt{21}},\ 0,\ \frac{1}{\sqrt{21}}\right)$.

Q1.14 [CALC] Integral Inner Product Evaluation

Source: *Spring 2023 FE Q5(a) [4 Marks]*

> Use the inner product $\langle \mathbf{f} \mid \mathbf{g} \rangle = \int_0^1 f(x)g(x)\,dx$ to compute $\langle \mathbf{f} \mid \mathbf{g} \rangle$ for $\mathbf{f} = \cos 2\pi x$ and $\mathbf{g} = \sin 2\pi x$.

Solution:

$$\langle \mathbf{f} \mid \mathbf{g} \rangle = \int_0^1 \cos(2\pi x)\sin(2\pi x)\,dx = \frac{1}{2}\int_0^1 \sin(4\pi x)\,dx = \frac{1}{2}\left[-\frac{\cos(4\pi x)}{4\pi}\right]_0^1 = -\frac{1}{8\pi}\big(\cos(4\pi) - \cos(0)\big) = -\frac{1}{8\pi}(1-1) = 0.$$

Part 2: General Vector Spaces (10 Axioms) & Properties

Q2.1 [PROOF] The Zero Vector Space $V = \{0\}$ (All 10 Axioms)

Source: *Spring 2024 Midterm Part B Q3 [10 Marks] (+ Official Sol)*

> Let V consist of a single object, denoted by $\mathbf{0}$, and define $\mathbf{0} + \mathbf{0} = \mathbf{0}$ and $k\mathbf{0} = \mathbf{0}$ for all scalars k. Show that V is a vector space.

Solution: Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$. Since $V = \{\mathbf{0}\}$, we necessarily have $\mathbf{u} = \mathbf{v} = \mathbf{w} = \mathbf{0}$. Verify all 10 axioms:

1. **Closure under addition:** $\mathbf{u} + \mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0} \in V$.
2. **Commutativity:** $\mathbf{u} + \mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{v} + \mathbf{u}$.
3. **Associativity:** $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = \mathbf{0} + (\mathbf{0} + \mathbf{0}) = \mathbf{0} + \mathbf{0} = (\mathbf{0} + \mathbf{0}) + \mathbf{0} = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$.
4. **Zero vector (Additive identity):** Since $\mathbf{0} \in V$, $\mathbf{0} + \mathbf{u} = \mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{0} = \mathbf{0} = \mathbf{u}$.
5. **Additive inverse:** Choose $-\mathbf{u} = \mathbf{0} \in V$; then $\mathbf{u} + (-\mathbf{u}) = (-\mathbf{u}) + \mathbf{u} = \mathbf{0} + \mathbf{0} = \mathbf{0}$.
6. **Closure under scalar multiplication:** For any $k \in \mathbb{R}$, $k\mathbf{u} = k\mathbf{0} = \mathbf{0} \in V$.
7. **Distributivity (scalar over vector sum):** $k(\mathbf{u} + \mathbf{v}) = k(\mathbf{0} + \mathbf{0}) = k\mathbf{0} = \mathbf{0} = \mathbf{0} + \mathbf{0} = k\mathbf{0} + k\mathbf{0} = k\mathbf{u} + k\mathbf{v}$.
8. **Distributivity (vector over scalar sum):** $(k + l)\mathbf{u} = (k + l)\mathbf{0} = \mathbf{0} = \mathbf{0} + \mathbf{0} = k\mathbf{0} + l\mathbf{0} = k\mathbf{u} + l\mathbf{u}$.
9. **Associativity of scalar multiplication:** $k(l\mathbf{u}) = k(l\mathbf{0}) = k\mathbf{0} = \mathbf{0} = (kl)\mathbf{0} = (kl)\mathbf{u}$.
10. **Multiplicative identity of 1:** $1\mathbf{u} = 1\mathbf{0} = \mathbf{0} = \mathbf{u}$.

All 10 axioms hold, so $V = \{\mathbf{0}\}$ is a vector space. ■

Q2.2 [PROOF] Positive Reals $V = (0, \infty)$ with $u \oplus v = uv$ and $k \odot u = u^k$

Sources:

- *Spring 2023 Midterm Q5 [10 Marks]* (Full proof of **all 10 axioms**)
- *Spring 2022 Midterm Q6(b) [5 Marks]* (Identify zero element & show Axiom 4)
- *Fall 2022 FE Q2(a) [6 Marks] & Spring 2024 FE Q3 [6 Marks]* (Identify $\mathbf{0}$ and $-\mathbf{u}$, prove Axioms 4 & 5)

> Consider the set $V = S = (0, \infty)$ of all positive real numbers with operations:
>
> - Vector addition: $\mathbf{u} \oplus \mathbf{v} = uv$ (ordinary multiplication)
> - Scalar multiplication: $k \odot \mathbf{u} = u^k$ (exponentiation, $k \in \mathbb{R}$)
>
> **(A)** Prove V is a vector space (all 10 axioms). **(B)** Identify the zero vector $\mathbf{0}$ and negative $-\mathbf{u}$, and verify Axioms 4 & 5.

Solution: Let $u, v, w \in (0, \infty)$ and $k, m \in \mathbb{R}$.

1. **Closure under addition:** Since $u > 0$ and $v > 0$, $u \oplus v = uv > 0$, so $u \oplus v \in V$.
2. **Commutativity:** $u \oplus v = uv = vu = v \oplus u$.
3. **Associativity:** $(u \oplus v) \oplus w = (uv)w = u(vw) = u \oplus (v \oplus w)$.
4. **Zero vector (Additive identity):** Choose $\mathbf{0}_V = 1$ (note $1 \in (0, \infty)$!). Check Axiom 4:
$$u \oplus \mathbf{0}_V = u \cdot 1 = u, \qquad \mathbf{0}_V \oplus u = 1 \cdot u = u.$$
5. **Additive inverse:** For each $u \in (0, \infty)$, define $-\mathbf{u} = \dfrac{1}{u} = u^{-1}$ (which is > 0, hence in V). Check Axiom 5:
$$u \oplus (-\mathbf{u}) = u \cdot \frac{1}{u} = 1 = \mathbf{0}_V, \qquad (-\mathbf{u}) \oplus u = \frac{1}{u} \cdot u = 1 = \mathbf{0}_V.$$
6. **Closure under scalar multiplication:** Since $u > 0$ and $k \in \mathbb{R}$, $k \odot u = u^k > 0$, so $k \odot u \in V$.
7. **Scalar distributivity over vector addition:**
$$k \odot (u \oplus v) = (uv)^k = u^k v^k = (k \odot u) \oplus (k \odot v).$$
8. **Vector distributivity over scalar addition:**
$$(k + m) \odot u = u^{k+m} = u^k u^m = (k \odot u) \oplus (m \odot u).$$
9. **Scalar associativity:**
$$k \odot (m \odot u) = k \odot (u^m) = (u^m)^k = u^{mk} = u^{km} = (km) \odot u.$$
10. **Scalar identity 1:**
$$1 \odot u = u^1 = u.$$

Hence $V = (0, \infty)$ is a vector space (and parts (B) follow directly from Axioms 4 & 5 with $\mathbf{0}_V = 1$ and $-\mathbf{u} = 1/u$). ■

Q2.3 [CHECK] Disproving a Candidate Vector Space & Finding ALL Failed Axioms

Source: *Fall 2022 Midterm Q6 [10 Marks]*

> Check whether the set $V = \mathbb{R}^3$ of all triples of real numbers with standard vector addition $\mathbf{u} + \mathbf{v} = (x_1 + x_2,\ y_1 + y_2,\ z_1 + z_2)$ and scalar multiplication defined by:
>
> $$k(x, y, z) = (k^2 x,\ k^2 y,\ k^2 z)$$
>
> is a vector space or not. If not, **identify all axioms that fail to hold**.

Solution: Let's systematically check all 10 axioms (for arbitrary $\mathbf{u} = (x_1, y_1, z_1), \mathbf{v} = (x_2, y_2, z_2) \in \mathbb{R}^3$ and $k, m \in \mathbb{R}$):

- **Axioms 1–5 (Addition only):** Standard addition on $\mathbb{R}^3$ is unchanged, so Axioms **1, 2, 3, 4, 5 all HOLD** ($\mathbf{0} = (0, 0, 0)$ and $-\mathbf{u} = (-x_1, -y_1, -z_1)$).
- **Axiom 6 (Closure under scalar multiplication):** $k(x, y, z) = (k^2 x, k^2 y, k^2 z) \in \mathbb{R}^3$. **HOLDS.**
- **Axiom 7 ($k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$):**
  - LHS: $k(x_1 + x_2, y_1 + y_2, z_1 + z_2) = (k^2(x_1 + x_2), k^2(y_1 + y_2), k^2(z_1 + z_2))$.
  - RHS: $(k^2 x_1, k^2 y_1, k^2 z_1) + (k^2 x_2, k^2 y_2, k^2 z_2) = (k^2 x_1 + k^2 x_2, k^2 y_1 + k^2 y_2, k^2 z_1 + k^2 z_2)$.
  - Since LHS $=$ RHS, Axiom **7 HOLDS.**
- **Axiom 8 ($(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$): FAILS!**
  - LHS: $(k + m)\mathbf{u} = \big((k + m)^2 x, (k + m)^2 y, (k + m)^2 z\big) = \big((k^2 + 2km + m^2)x, \dots\big)$.
  - RHS: $k\mathbf{u} + m\mathbf{u} = \big((k^2 + m^2)x, (k^2 + m^2)y, (k^2 + m^2)z\big)$.
  - Counterexample: take $k = 1, m = 1, \mathbf{u} = (1, 0, 0)$: LHS $= 2(1, 0, 0) = (4, 0, 0)$, whereas RHS $= 1(1, 0, 0) + 1(1, 0, 0) = (1, 0, 0) + (1, 0, 0) = (2, 0, 0) \neq (4, 0, 0)$.

- **Axiom 9 ($k(m\mathbf{u}) = (km)\mathbf{u}$):**
  - LHS: $k(m^2x, m^2y, m^2z) = (k^2m^2x, k^2m^2y, k^2m^2z)$.
  - RHS: $((km)^2x, (km)^2y, (km)^2z) = (k^2m^2x, k^2m^2y, k^2m^2z)$.
  - Wait — Axiom **9 HOLDS!**
- **Axiom 10 ($1\mathbf{u} = \mathbf{u}$):**
  - $1(x, y, z) = (1^2x, 1^2y, 1^2z) = (x, y, z)$. Axiom **10 HOLDS.**

Conclusion: V is **NOT** a vector space. **Only Axiom 8 fails** (Axioms 1–7, 9, and 10 all hold).

Q2.4 [PROOF] Can a Vector Space Contain Only Two Distinct Elements?

Source: *Spring 2022 Midterm Q6(a) [5 Marks]*

> Prove or disprove the following claim: *"It is possible for a vector space (over $\mathbb{R}$) to contain only two distinct elements."*

Solution: The claim is **FALSE** (disprove by contradiction). Suppose V is a real vector space with $|V| = 2$. Since every vector space contains a zero vector $\mathbf{0}$, there must exist some nonzero vector $\mathbf{v} \in V$ ($\mathbf{v} \neq \mathbf{0}$), so $V = \{\mathbf{0}, \mathbf{v}\}$. By Axiom 6 (closure under scalar multiplication by any real scalar $k \in \mathbb{R}$), both $2\mathbf{v}$ and $\mathbf{v}$ must belong to V, and in particular $2\mathbf{v} \in \{\mathbf{0}, \mathbf{v}\}$:

- If $2\mathbf{v} = \mathbf{0}$, multiply both sides by $\frac{1}{2}$: $\frac{1}{2}(2\mathbf{v}) = \frac{1}{2}\mathbf{0} \implies \mathbf{v} = \mathbf{0}$, contradicting $\mathbf{v} \neq \mathbf{0}$.
- If $2\mathbf{v} = \mathbf{v}$, add $-\mathbf{v}$ to both sides: $2\mathbf{v} - \mathbf{v} = \mathbf{0} \implies (2-1)\mathbf{v} = \mathbf{0} \implies \mathbf{v} = \mathbf{0}$, again contradicting $\mathbf{v} \neq \mathbf{0}$.

Hence $2\mathbf{v}$ cannot equal either $\mathbf{0}$ or $\mathbf{v}$, so V must contain infinitely many distinct elements ($k\mathbf{v}$ for all $k \in \mathbb{R}$ are distinct). A real vector space can only have 1 element ($V = \{\mathbf{0}\}$) or infinitely many elements—never 2. ■

Q2.5 [PROOF] Fundamental Arithmetic Properties in Any Vector Space

Source: *Fall 2022 FE Q3(a),(b) [5+5 Marks]*

> Let V be a vector space, $\mathbf{u} \in V$, and k a scalar. Prove:
>
> - **(a)** $k\mathbf{0} = \mathbf{0}$
> - **(b)** $(-1)\mathbf{u} = -\mathbf{u}$

Solution:

- **(a) Prove $k\mathbf{0} = \mathbf{0}$:** By Axiom 4, $\mathbf{0} + \mathbf{0} = \mathbf{0}$. Multiplying both sides by k and applying Axiom 7:

  $$k\mathbf{0} = k(\mathbf{0} + \mathbf{0}) = k\mathbf{0} + k\mathbf{0}.$$

  Add the additive inverse $-(k\mathbf{0})$ (which exists by Axiom 5) to both sides:

  $$k\mathbf{0} + \big(-(k\mathbf{0})\big) = (k\mathbf{0} + k\mathbf{0}) + \big(-(k\mathbf{0})\big).$$

  By Axioms 3, 5, and 4:

  $$\mathbf{0} = k\mathbf{0} + \Big(k\mathbf{0} + \big(-(k\mathbf{0})\big)\Big) = k\mathbf{0} + \mathbf{0} = k\mathbf{0}. \quad \blacksquare$$

- **(b) Prove $(-1)\mathbf{u} = -\mathbf{u}$:** We must show that $(-1)\mathbf{u}$ satisfies the defining property of the additive inverse of $\mathbf{u}$, namely $\mathbf{u} + (-1)\mathbf{u} = \mathbf{0}$:

  $$\begin{aligned} \mathbf{u} + (-1)\mathbf{u} &= 1\mathbf{u} + (-1)\mathbf{u} && (\text{Axiom } 10) \\ &= \big(1 + (-1)\big)\mathbf{u} && (\text{Axiom } 8) \\ &= 0\mathbf{u} \\ &= \mathbf{0} && (\text{since } 0\mathbf{u} = (0+0)\mathbf{u} = 0\mathbf{u} + 0\mathbf{u} \implies 0\mathbf{u} = \mathbf{0}). \end{aligned}$$

  By Axiom 2, $(-1)\mathbf{u} + \mathbf{u} = \mathbf{0}$ as well. Since additive inverses are unique (add $-\mathbf{u}$ to $\mathbf{u} + (-1)\mathbf{u} = \mathbf{0}$), it follows that $(-1)\mathbf{u} = -\mathbf{u}$. ■

Part 3: Subspaces & The Subspace Test

Q3.1 [PROOF] The Two-Condition Subspace Test

Sources: *Fall 2022 Midterm Q7 [10 Marks]; Spring 2024 FE Q5 [10 Marks]*

> **Prove:** If W is a set of one or more vectors from a vector space V, then W is a subspace of V if and only if:
>
> - **(a)** If $\mathbf{u}, \mathbf{v} \in W$, then $\mathbf{u} + \mathbf{v} \in W$.
> - **(b)** If k is any scalar and $\mathbf{u} \in W$, then $k\mathbf{u} \in W$.

Solution:

- **($\Rightarrow$) Forward Direction:** Assume W is a subspace of V. By definition, W is itself a vector space under the addition and scalar multiplication of V. Therefore, it satisfies **Axiom 1** (closure under addition, which is condition (a)) and **Axiom 6** (closure under scalar multiplication, which is condition (b)).
- **($\Leftarrow$) Converse Direction:** Assume W is nonempty and conditions (a) and (b) hold. To show W is a subspace of V, we must verify all 10 vector space axioms for W:
  - **Axiom 1** holds by condition (a).
  - **Axiom 6** holds by condition (b).
  - **Axioms 2, 3, 7, 8, 9, 10** hold automatically for all vectors in W because every vector in W also belongs to the parent vector space V, where these equality-based algebraic laws already hold.
  - **Axiom 4 (Zero vector $\mathbf{0} \in W$):** Since W has at least one vector, pick any $\mathbf{u} \in W$. Setting $k = 0$ in condition (b) gives $0\mathbf{u} \in W$. Since $0\mathbf{u} = \mathbf{0}$ in V, we have $\mathbf{0} \in W$ (and $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$).
  - **Axiom 5 (Additive inverses in W):** For any $\mathbf{u} \in W$, setting $k = -1$ in condition (b) gives $(-1)\mathbf{u} \in W$. Since $(-1)\mathbf{u} = -\mathbf{u}$ in V, we have $-\mathbf{u} \in W$ (and $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$).

  Since all 10 axioms hold in W, W is a subspace of V. ■

Q3.2 [T/F + PROOF] One-Step Subspace Test & Intersection of Two Subspaces

Sources: *Fall 2022 Mid Q8(a)(i); Spring 2022 FE Q4(a), Q4(e)*

> Indicate whether each statement is **always true** or **sometimes false**, and justify:
>
> - **(a)** *The intersection of two subspaces of a vector space V is also a subspace of V. (Sp22 FE Q4a)*
> - **(b)** *If W is a nonempty set of vectors from a vector space V, and $k\mathbf{u} + \mathbf{v} \in W$ for all $\mathbf{u}, \mathbf{v} \in W$ and all scalars $k \in \mathbb{R}$, then W is a subspace of V. (F22 Mid Q8a(i), Sp22 FE Q4e)*

Solution:

- **(a) ALWAYS TRUE.** Let W_1, W_2 be subspaces of V, and $W = W_1 \cap W_2$.
  1. Since $\mathbf{0} \in W_1$ and $\mathbf{0} \in W_2$, we have $\mathbf{0} \in W$, so $W \neq \emptyset$.
  2. If $\mathbf{u}, \mathbf{v} \in W$, then $\mathbf{u}, \mathbf{v} \in W_1$ (so $\mathbf{u} + \mathbf{v} \in W_1$) and $\mathbf{u}, \mathbf{v} \in W_2$ (so $\mathbf{u} + \mathbf{v} \in W_2$). Hence $\mathbf{u} + \mathbf{v} \in W_1 \cap W_2 = W$.
  3. If $k \in \mathbb{R}$ and $\mathbf{u} \in W$, then $k\mathbf{u} \in W_1$ and $k\mathbf{u} \in W_2$, so $k\mathbf{u} \in W$. By the Subspace Test, $W_1 \cap W_2$ is a subspace of V.
- **(b) ALWAYS TRUE.** Given $W \neq \emptyset$ and $k\mathbf{u} + \mathbf{v} \in W$ for all $\mathbf{u}, \mathbf{v} \in W, k \in \mathbb{R}$:
  1. **Closed under addition:** Set $k = 1$; then $1\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{v} \in W$ for all $\mathbf{u}, \mathbf{v} \in W$.
  2. **Zero vector is in W:** Pick any $\mathbf{u}_0 \in W$ (exists since $W \neq \emptyset$) and set $\mathbf{u} = \mathbf{v} = \mathbf{u}_0, k = -1$; then $(-1)\mathbf{u}_0 + \mathbf{u}_0 = \mathbf{0} \in W$.
  3. **Closed under scalar multiplication:** For any $\mathbf{u} \in W$ and $k \in \mathbb{R}$, set $\mathbf{v} = \mathbf{0} \in W$; then $k\mathbf{u} + \mathbf{0} = k\mathbf{u} \in W$. By the Subspace Test, W is a subspace of V.

Q3.3 [PROOF + CALC] Subspaces of Polynomials P_n

Sources: *Spring 2025 FE Solution P9(a),(b); Spring 2022 Midterm Q10 [10 Marks]*

> - **(a) (Sp25 FE P9a):** Show that $W_1 = \{p(x) \in P_n : p(-x) = p(x)\}$ (even polynomials) is a subspace of P_n.
> - **(b) (Sp25 FE P9b):** Show that $W_2 = \{p(x) \in P_n : p'(0) = 0\}$ is a subspace of P_n, and find a basis for W_2.
> - **(c) (Sp22 Mid Q10):** Show that the set W_3 of all polynomials in P_n that have a horizontal tangent at $x = 1$ (i.e., $p'(1) = 0$) is a subspace of P_n.

Solution:

- **(a)** Let $W_1 = \{p \in P_n : p(-x) = p(x)\}$:
  1. **Zero polynomial:** $0(-x) = 0 = 0(x) \implies 0 \in W_1$ ($W_1 \neq \emptyset$).
  2. **Addition:** If $p, q \in W_1$, then $(p+q)(-x) = p(-x) + q(-x) = p(x) + q(x) = (p+q)(x) \implies p+q \in W_1$.
  3. **Scalar mult:** If $p \in W_1, c \in \mathbb{R}$, then $(cp)(-x) = c\,p(-x) = c\,p(x) = (cp)(x) \implies cp \in W_1$. Hence W_1 is a subspace of P_n. ■
- **(b)** Let $W_2 = \{p \in P_n : p'(0) = 0\}$:
  1. **Zero polynomial:** $0'(x) = 0 \implies 0'(0) = 0 \implies 0 \in W_2$.
  2. **Addition:** If $p, q \in W_2$, $(p+q)'(0) = p'(0) + q'(0) = 0 + 0 = 0 \implies p+q \in W_2$.
  3. **Scalar mult:** If $p \in W_2, c \in \mathbb{R}$, $(cp)'(0) = c\,p'(0) = c \cdot 0 = 0 \implies cp \in W_2$. Thus W_2 is a subspace of P_n.
  - **Basis (for $n \geq 2$; if $n = 0, 1$ just $\{1\}$):** Any $p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \in P_n$ has $p'(x) = a_1 + 2a_2 x + \cdots + n a_n x^{n-1}$, so $p'(0) = a_1$. Thus $p \in W_2 \iff a_1 = 0 \iff p(x) = a_0 + a_2 x^2 + \cdots + a_n x^n$. Since $\{1, x^2, x^3, \ldots, x^n\}$ spans W_2 and is a subset of the linearly independent standard basis $\{1, x, \ldots, x^n\}$, it is a **basis for** W_2:

$$\text{Basis for } W_2 = \{1, x^2, x^3, \ldots, x^n\} \quad (\text{or } \{1\} \text{ if } n \leq 1).$$

- **(c)** Horizontal tangent at $x = 1$ means $p'(1) = 0$, so $W_3 = \{p \in P_n : p'(1) = 0\}$.
  1. **Zero polynomial:** $p(x) = 0 \implies p'(x) = 0 \implies p'(1) = 0 \implies 0 \in W_3$ ($W_3 \neq \emptyset$).
  2. **Addition:** Let $p, q \in W_3$, so $p'(1) = 0$ and $q'(1) = 0$. By linearity of differentiation, $(p+q)'(1) = p'(1) + q'(1) = 0 + 0 = 0 \implies p+q \in W_3$.
  3. **Scalar mult:** Let $p \in W_3$ and $k \in \mathbb{R}$. Then $(kp)'(1) = k\,p'(1) = k(0) = 0 \implies kp \in W_3$. By the Subspace Test, W_3 is a subspace of P_n. ■

Q3.4 [PROOF / COUNTEREXAMPLE] Matrix Subspaces in $M_{3\times3}(\mathbb{R})$

Source: *Quiz 6 Problems 1, 4, 5, 6 (+ Official Sol)*

> Determine whether each subset W of $V = M_{3\times3}(\mathbb{R})$ is a subspace of V. Justify.
>
> - **(a) (P1):** $W =$ set of all 3×3 real **magic squares** (every row, column, and both main diagonals sum to the same number).
> - **(b) (P4):** $W = \{A \in V \mid \det(A) = 0\}$.
> - **(c) (P5):** $W = \{A \in V \mid A^T = A\}$ (symmetric matrices).
> - **(d) (P6):** $W = \{A \in V \mid A^T = -A\}$ (skew-symmetric matrices).

Solution:

- **(a) YES (Subspace):**
  1. Zero matrix $O = [0]_{3\times3}$ has all row, column, and diagonal sums equal to $0 \implies O \in W$.
  2. Let $A \in W$ have magic sum s_1 and $B \in W$ have magic sum s_2. Then every row sum of $A + B$ is $\sum_{j=1}^{3}(a_{ij} + b_{ij}) = s_1 + s_2$, every column sum is $\sum_{i=1}^{3}(a_{ij} + b_{ij}) = s_1 + s_2$, and both diagonal sums are also $s_1 + s_2$. Hence $A + B \in W$.
  3. For $k \in \mathbb{R}$, every row, column, and diagonal sum of kA equals ks_1, so $kA \in W$. Thus W is a subspace of $M_{3\times3}(\mathbb{R})$.
- **(b) NO (Not a subspace):** While $O \in W$ ($\det(O) = 0$) and W is closed under scalar multiplication ($\det(kA) = k^3 \det(A) = 0$), W **is NOT closed under addition**: Take $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ ($\det A = 0$) and $B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ($\det B = 0$). Both $A, B \in W$, yet $A + B = I_3$, and $\det(I_3) = 1 \neq 0$, so $A + B \notin W$. *(Note: Quiz 6 key wrote a tiny typo $0 \cdot 1 \cdot 1$ instead of $1 \cdot 1 \cdot 0$ for $\det A$; fixed here.)*
- **(c) YES (Subspace):**
  1. $O^T = O \implies O \in W$.
  2. If $A, B \in W$, $(A+B)^T = A^T + B^T = A + B \implies A + B \in W$.
  3. If $A \in W, k \in \mathbb{R}$, $(kA)^T = kA^T = kA \implies kA \in W$.
- **(d) YES (Subspace):**

1. $O^T = O = -O \implies O \in W$.
2. If $A, B \in W$, $(A+B)^T = A^T + B^T = (-A) + (-B) = -(A+B) \implies A+B \in W$.
3. If $A \in W, k \in \mathbb{R}$, $(kA)^T = kA^T = k(-A) = -(kA) \implies kA \in W$.

Q3.5 [PROOF] Function Subspace via Integral Condition

Source: *Quiz 6 Problem 7 (+ Official Sol)*

> Let $V = C([a,b])$ be all continuous functions on $[a,b]$. Is $W = \left\{ f \in V \,\middle|\, \int_a^b f(x)\,dx = 0 \right\}$ a subspace of V? Justify.

Solution: YES (Subspace).

1. Zero function $0(x) = 0$ satisfies $\int_a^b 0\,dx = 0 \implies 0 \in W$.
2. If $f, g \in W$, $\int_a^b (f(x)+g(x))\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx = 0 + 0 = 0 \implies f+g \in W$.
3. If $f \in W, k \in \mathbb{R}$, $\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx = k \cdot 0 = 0 \implies kf \in W$. ■

Part 4: Span, Linear Independence, Bases & Dimension

Q4.1 [PROOF] $\mathrm{Span}\{v_1, \ldots, v_r\}$ Is the Smallest Subspace

Source: *Fall 2022 FE Q8(a),(b) [5+5 Marks]*

> If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r$ are vectors in a vector space U, prove:
>
> - **(a)** The set W of all linear combinations of $\mathbf{v}_1, \ldots, \mathbf{v}_r$ is a subspace of U.
> - **(b)** W is the smallest subspace of U that contains $\mathbf{v}_1, \ldots, \mathbf{v}_r$.

Solution:

- **(a) W is a subspace of U:**
  - *Nonempty:* $\mathbf{0} = 0\mathbf{v}_1 + \cdots + 0\mathbf{v}_r \in W$ (and each $\mathbf{v}_i \in W$).
  - *Addition:* Let $\mathbf{x} = \sum_{i=1}^r c_i \mathbf{v}_i \in W$ and $\mathbf{y} = \sum_{i=1}^r d_i \mathbf{v}_i \in W$. Then:

$$\mathbf{x} + \mathbf{y} = \sum_{i=1}^{r} (c_i + d_i)\mathbf{v}_i \in W.$$

  - *Scalar multiplication:* For any $k \in \mathbb{R}$,

$$k\mathbf{x} = \sum_{i=1}^{r} (kc_i)\mathbf{v}_i \in W.$$

  By the Subspace Test, W is a subspace of U.

- **(b) W is the smallest subspace containing $\mathbf{v}_1, \ldots, \mathbf{v}_r$:** First, W contains every $\mathbf{v}_i$ because $\mathbf{v}_i = 0\mathbf{v}_1 + \cdots + 1\mathbf{v}_i + \cdots + 0\mathbf{v}_r$. Next, let W' be *any* subspace of U that contains $\mathbf{v}_1, \ldots, \mathbf{v}_r$. Because W' is closed under scalar multiplication and addition, W' must contain every linear combination $c_1\mathbf{v}_1 + \cdots + c_r\mathbf{v}_r$. Thus $W \subseteq W'$. Therefore W is contained in every such subspace, making it the smallest. ■

Q4.2 [CONCEPT / CALC] Trigonometric Span & Spanning Matrices for M_{nn}

Source: *Fall 2022 Midterm Q8(b), Q10 [10 Marks]*

> - **(a) (F22 Mid Q10):** Let V be the subspace of $F(-\infty, \infty)$ given by $V = \mathrm{span}\{\sin x, \cos x\}$. Show that for any given value of θ, $\mathbf{f}_1 = \cos(x+\theta)$ and $\mathbf{g}_1 = \sin(x+\theta)$ belong to V.
> - **(b) (F22 Mid Q8b):** In words, describe a set of matrices that spans M_{nn}.

Solution:

- **(a)** Using the angle-addition identities:

$$\mathbf{f}_1(x) = \cos(x+\theta) = (\cos\theta)\cos x + (-\sin\theta)\sin x,$$
$$\mathbf{g}_1(x) = \sin(x+\theta) = (\cos\theta)\sin x + (\sin\theta)\cos x.$$

Since θ is fixed, $\cos\theta$ and $\sin\theta$ are constants (scalars). Both $\mathbf{f}_1$ and $\mathbf{g}_1$ are expressed as linear combinations of $\sin x$ and $\cos x$, so $\mathbf{f}_1 \in V$ and $\mathbf{g}_1 \in V$. ■

- **(b)** The standard basis set of n^2 matrices E_{ij} ($1 \le i, j \le n$), where **each matrix E_{ij} has 1 in the (i,j)-th entry and 0 in all other entries**, spans M_{nn} (since any $n \times n$ matrix $A = [a_{ij}]$ is uniquely written as $\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}E_{ij}$).

Q4.3 [CALC] Linear Combinations & Spanning Sets in $\mathbb{R}^2$ and $\mathbb{R}^3$

Source: *Quiz 7 Problems 1–8 (+ Official Sol)*

> Solve each linear-combination / spanning problem:
>
> 1. **(P1)** Find scalars a, b such that $a(3,2) + b(-1,4) = (7,-6)$.
> 2. **(P2)** Do $\mathbf{v} = (1,2,3), \mathbf{w} = (2,3,4), \mathbf{u} = (3,4,5)$ span $\mathbb{R}^3$?
> 3. **(P3)** Find scalars a, b such that $a(2,1) + b(-1,3) = (5,0)$.
> 4. **(P4)** Can $\mathbf{u} = (4,-2)$ be expressed as a linear combination of $\mathbf{v} = (1,2)$ and $\mathbf{w} = (3,-1)$?
> 5. **(P5)** Do $\mathbf{v} = (1,0,1), \mathbf{w} = (0,1,1), \mathbf{u} = (1,1,0)$ span $\mathbb{R}^3$?
> 6. **(P6)** Can $\mathbf{u} = (6,-3)$ be expressed as a linear combination of $\mathbf{v} = (1,2)$ and $\mathbf{w} = (2,-1)$?
> 7. **(P7)** Can $\mathbf{u} = (5,1)$ be expressed as a linear combination of $\mathbf{v} = (2,1)$ and $\mathbf{w} = (4,2)$?
> 8. **(P8)** Do $\mathbf{v} = (1,1,0), \mathbf{w} = (0,1,1), \mathbf{u} = (1,0,1)$ span $\mathbb{R}^3$?

Solutions:

1. **(P1):** $3a - b = 7$ and $2a + 4b = -6 \implies 14a = 22 \implies \mathbf{a = \frac{11}{7}},\ \mathbf{b = -\frac{16}{7}}$.
2. **(P2):** Put vectors as columns of $A = \begin{bmatrix}1&2&3\\2&3&4\\3&4&5\end{bmatrix}$. $\det(A) = 1(15-16) - 2(10-12) + 3(8-9) = -1 + 4 - 3 = 0$.
   Since $\det(A) = 0$ (note $\mathbf{v} - 2\mathbf{w} + \mathbf{u} = \mathbf{0}$), **No, they do not span $\mathbb{R}^3$**.
3. **(P3):** $2a - b = 5$ and $a + 3b = 0 \implies a = -3b \implies -7b = 5 \implies \mathbf{a = \frac{15}{7}},\ \mathbf{b = -\frac{5}{7}}$.
4. **(P4):** Solve $a + 3b = 4,\ 2a - b = -2 \implies 7a = -2 \implies a = -\frac{2}{7},\ b = \frac{10}{7}$. Solution exists $\implies$ **Yes**.
5. **(P5):** $A = \begin{bmatrix}1&0&1\\0&1&1\\1&1&0\end{bmatrix}$, $\det(A) = 1(0-1) + 1(0-1) = -2 \neq 0 \implies$ **Yes, they span $\mathbb{R}^3$** (in fact a basis).
6. **(P6):** Solve $a + 2b = 6,\ 2a - b = -3 \implies 5a = 0 \implies a = 0, b = 3$. Solution exists $\implies$ **Yes**.
7. **(P7):** Solve $2a + 4b = 5,\ a + 2b = 1$. Multiplying the 2nd by 2 gives $2a + 4b = 2 \neq 5$ ($0 = 3$, contradiction) $\implies$ **No**.
8. **(P8):** $A = \begin{bmatrix}1&0&1\\1&1&0\\0&1&1\end{bmatrix}$, $\det(A) = 1(1-0) + 1(1-0) = 2 \neq 0 \implies$ **Yes, they span $\mathbb{R}^3$**.

Q4.4 [T/F] Concept Checks on Span and Sets Containing $\mathbf{0}$

Sources: *Fall 2022 Mid Q8(a)(ii); Spring 2022 FE Q4(b), Q4(d)*

> Always True or Sometimes False? Justify:
>
> - **(a)** *If $\text{span}(S_1) = \text{span}(S_2)$, then $S_1 = S_2$. (F22 Mid Q8a(ii), Sp22 FE Q4d)*
> - **(b)** *A set that contains the zero vector is linearly dependent. (Sp22 FE Q4b)*

Solution:

- **(a) SOMETIMES FALSE.** Counterexample in $\mathbb{R}^2$: $S_1 = \{(1,0),(0,1)\}$ and $S_2 = \{(1,0),(0,1),(1,1)\}$ both satisfy $\text{span}(S_1) = \text{span}(S_2) = \mathbb{R}^2$, even though $S_1 \neq S_2$.
- **(b) ALWAYS TRUE.** Suppose $S = \{\mathbf{0}, \mathbf{v}_1, \ldots, \mathbf{v}_r\}$. Then the nontrivial linear combination:

  $$1 \cdot \mathbf{0} + 0\mathbf{v}_1 + \cdots + 0\mathbf{v}_r = \mathbf{0}$$

  has a nonzero coefficient ($1 \neq 0$), so S is linearly dependent.

Q4.5 [PROOF] Two-Vector Independence Criterion & Difference Vectors

Sources: *Fall 2022 FE Q2(b) [4 Marks]; Spring 2022 Mid Q7(a) [5 Marks]*

> - **(a) (F22 FE Q2b):** Prove: A set with exactly two vectors $\{\mathbf{u}, \mathbf{v}\}$ is linearly independent if and only if neither vector is a scalar multiple of the other.
> - **(b) (Sp22 Mid Q7a):** Prove that for any vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$, the vectors $(\mathbf{a}-\mathbf{b})$, $(\mathbf{b}-\mathbf{c})$, and $(\mathbf{c}-\mathbf{a})$ form a linearly dependent set.

Solution:

- **(a)** Equivalently prove the contrapositive: $\{\mathbf{u}, \mathbf{v}\}$ is linearly dependent $\iff$ one is a scalar multiple of the other.
  - *($\Rightarrow$)* Suppose $\{\mathbf{u}, \mathbf{v}\}$ is LD. Then $c_1\mathbf{u} + c_2\mathbf{v} = \mathbf{0}$ for scalars c_1, c_2 not both 0. If $c_1 \neq 0$, then $\mathbf{u} = \left(-\frac{c_2}{c_1}\right)\mathbf{v}$; if $c_2 \neq 0$, then $\mathbf{v} = \left(-\frac{c_1}{c_2}\right)\mathbf{u}$. Either way, one is a scalar multiple of the other.
  - *($\Leftarrow$)* Suppose without loss of generality $\mathbf{u} = k\mathbf{v}$ for some $k \in \mathbb{R}$. Then $1 \cdot \mathbf{u} + (-k)\mathbf{v} = \mathbf{0}$ with coefficient $1 \neq 0$, so $\{\mathbf{u}, \mathbf{v}\}$ is LD. ■
- **(b)** Consider the linear combination with coefficients $c_1 = 1,\ c_2 = 1,\ c_3 = 1$:

$$1(\mathbf{a}-\mathbf{b}) + 1(\mathbf{b}-\mathbf{c}) + 1(\mathbf{c}-\mathbf{a}) = (\mathbf{a}-\mathbf{a}) + (-\mathbf{b}+\mathbf{b}) + (-\mathbf{c}+\mathbf{c}) = \mathbf{0}.$$

  Since $(1,1,1) \neq (0,0,0)$ is a nontrivial solution, the set $\{\mathbf{a}-\mathbf{b},\ \mathbf{b}-\mathbf{c},\ \mathbf{c}-\mathbf{a}\}$ is linearly dependent. ■

Q4.6 [PROOF] Adding a Vector Outside Span & General Linear Independence Equivalence

Source: *Spring 2022 Midterm Q7(b) [5 Marks], Q8 [5+5 Marks]*

> - **(a) (Sp22 Mid Q7b):** Prove that if $\{\mathbf{a}, \mathbf{b}\}$ is a linearly independent set, and $\mathbf{c} \notin \text{span}\{\mathbf{a}, \mathbf{b}\}$, then $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ is also linearly independent.
> - **(b) (Sp22 Mid Q8):** Given a set S with two or more vectors:
>   - (i) Prove that if S is linearly independent, then no vector in S is expressible as a linear combination of the other vectors in S.
>   - (ii) Prove that if no vector in S is expressible as a linear combination of the other vectors in S, then S is linearly independent.

Solution:

- **(a)** Suppose $k_1\mathbf{a} + k_2\mathbf{b} + k_3\mathbf{c} = \mathbf{0}$.
  - If $k_3 \neq 0$, we could divide by k_3 to get $\mathbf{c} = \left(-\frac{k_1}{k_3}\right)\mathbf{a} + \left(-\frac{k_2}{k_3}\right)\mathbf{b} \in \text{span}\{\mathbf{a}, \mathbf{b}\}$, contradicting $\mathbf{c} \notin \text{span}\{\mathbf{a}, \mathbf{b}\}$.
  - Therefore $k_3 = 0$, leaving $k_1\mathbf{a} + k_2\mathbf{b} = \mathbf{0}$. Because $\{\mathbf{a}, \mathbf{b}\}$ is linearly independent, $k_1 = k_2 = 0$. Thus $k_1 = k_2 = k_3 = 0$, so $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ is linearly independent. ■
- **(b) General theorem (proved for finite $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$, $r \geq 2$ — by contrapositive of each direction):**
  - *(i) ($\Rightarrow$):* Assume S is LI. Suppose for contradiction some $\mathbf{v}_i$ is a linear combination of the others: $\mathbf{v}_i = \sum_{j \neq i} c_j\mathbf{v}_j$. Moving $\mathbf{v}_i$ to the RHS gives:

$$c_1\mathbf{v}_1 + \cdots + (-1)\mathbf{v}_i + \cdots + c_r\mathbf{v}_r = \mathbf{0},$$

    which has coefficient $-1 \neq 0$, contradicting that S is LI. Thus no vector in S can be a linear combination of the others.
  - *(ii) ($\Leftarrow$):* Prove contrapositive: suppose S is linearly dependent. Then there exist scalars $k_1, \ldots, k_r$, not all zero, such that $k_1\mathbf{v}_1 + \cdots + k_r\mathbf{v}_r = \mathbf{0}$. Pick any index i with $k_i \neq 0$ and solve for $\mathbf{v}_i$:

$$\mathbf{v}_i = \sum_{j \neq i} \left(-\frac{k_j}{k_i}\right)\mathbf{v}_j,$$

    showing that $\mathbf{v}_i$ IS expressible as a linear combination of the other vectors in S. Contrapositively, if no vector is a linear combination of the others, S must be linearly independent. ■

Q4.7 [PROOF] Orthogonal Set of Nonzero Vectors Is Linearly Independent

Sources: *Spring 2023 FE Q8 [10 Marks]; Quiz 11 Problem 4(a) (+ Official Sol)*

> Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be an orthogonal set of nonzero vectors in an inner product space V. Prove that S is linearly independent.

Solution: Suppose $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$. Fix any $k \in \{1, \ldots, n\}$ and take the inner product of both sides with $\mathbf{v}_k$:

$$\langle c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k + \cdots + c_n\mathbf{v}_n,\ \mathbf{v}_k\rangle = \langle \mathbf{0}, \mathbf{v}_k\rangle = 0.$$

By linearity and orthogonality ($\langle \mathbf{v}_i, \mathbf{v}_k\rangle = 0$ for all $i \neq k$), every term vanishes except the k-th:

$$c_k\langle \mathbf{v}_k, \mathbf{v}_k\rangle = 0 \implies c_k\|\mathbf{v}_k\|^2 = 0.$$

Since $\mathbf{v}_k \neq \mathbf{0}$, $\|\mathbf{v}_k\|^2 > 0$, forcing $c_k = 0$. Since k was arbitrary, $c_1 = c_2 = \cdots = c_n = 0$, so S is linearly independent. ■

Q4.8 [PROOF] Size Bounds: $> n$ Vectors Is LD & $< n$ Vectors Doesn't Span

Sources:

- *Spring 2023 Mid Q6 [10 Marks] & Spring 2023 FE Q4 [10 Marks]* ($S = \{\mathbf{v}_1, \ldots, \mathbf{v}_r\} \subset \mathbb{R}^n, r > n \implies S$ is not LI)
- *Spring 2024 FE Q2 [8 Marks] & Spring 2025 FE Solution P3(a),(b)* (General finite-dim V with basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$)

> Let V be a finite-dimensional vector space with basis $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ (or $V = \mathbb{R}^n$ directly). Prove:
>
> - **(a)** Any set $S' = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$ with $m > n$ vectors is linearly dependent.
> - **(b)** Any set $S' = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$ with $m < n$ vectors does not span V.

Solution:

- **(a) Case $V = \mathbb{R}^n$ (Spring 23 Mid/FE):** Consider $k_1\mathbf{w}_1 + \cdots + k_m\mathbf{w}_m = \mathbf{0}$ with $\mathbf{w}_j = (w_{1j}, \ldots, w_{nj})^T$. Equating components gives a **homogeneous linear system of n equations in m unknowns** ($k_1, \ldots, k_m$). Since $m > n$ (more unknowns than equations), the system has free variable(s) and hence a **nontrivial solution**. Thus $\{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$ is linearly dependent.

- **(a) General V (Spring 24 FE / Spring 25 FE P3a):** Since $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis for V, each $\mathbf{w}_j$ can be written as $\mathbf{w}_j = \sum_{i=1}^{n} a_{ij}\mathbf{v}_i$. Substitute into $k_1\mathbf{w}_1 + \cdots + k_m\mathbf{w}_m = \mathbf{0}$:

  $$\sum_{j=1}^{m} k_j \left(\sum_{i=1}^{n} a_{ij}\mathbf{v}_i\right) = \sum_{i=1}^{n} \left(\sum_{j=1}^{m} a_{ij}k_j\right)\mathbf{v}_i = \mathbf{0}.$$

  Because $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly independent, every coefficient must be 0:

  $$\sum_{j=1}^{m} a_{ij}k_j = 0 \qquad (i = 1, 2, \ldots, n).$$

  This is a homogeneous system of n equations in $m > n$ unknowns $k_1, \ldots, k_m$, which always has a **nontrivial solution**. Hence S' is linearly dependent. ■

- **(b) General V (Spring 25 FE P3b):** Suppose for contradiction that $S' = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$ ($m < n$) spans V. Then each basis vector $\mathbf{v}_j \in S$ ($j = 1, \ldots, n$) can be written as $\mathbf{v}_j = \sum_{i=1}^{m} a_{ij}\mathbf{w}_i$. Consider the equation $k_1\mathbf{v}_1 + \cdots + k_n\mathbf{v}_n = \mathbf{0}$:

  $$\sum_{j=1}^{n} k_j \left(\sum_{i=1}^{m} a_{ij}\mathbf{w}_i\right) = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij}k_j\right)\mathbf{w}_i = \mathbf{0}.$$

  Setting each coefficient of $\mathbf{w}_i$ equal to zero yields a homogeneous system $\sum_{j=1}^{n} a_{ij}k_j = 0$ ($i = 1, \ldots, m$) of m **equations in $n > m$ unknowns**. This system has a nontrivial solution $(k_1, \ldots, k_n) \neq \mathbf{0}$, which gives a nontrivial dependence $\sum_{j=1}^{n} k_j\mathbf{v}_j = \mathbf{0}$, contradicting that the basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly independent. Therefore S' cannot span V. ■

Q4.9 [PROOF + FILL-IN] The Plus / Minus Basis Theorems

Sources: *Fall 2022 FE Q4(a),(b) [5+5 Marks]; Spring 2022 Midterm Q9 [10 Marks]*

> Let S be a finite set of vectors in a finite-dimensional vector space V ($V \neq \{\mathbf{0}\}$):

> - **(a) (Minus Theorem):** If S spans V but is not a basis for V, then S can be **reduced** to a basis for V by **removing** appropriate vectors from S. *(Also Sp22 Mid Q9: S finite, $\text{span}(S) = V$, S LD $\implies$ S has a proper subset $B \subset S$ that is a basis for V.)*
> - **(b) (Plus Theorem):** If S is a linearly independent set that is not already a basis for V, then S can be **enlarged** (or **extended**) to a basis for V by **inserting** appropriate vectors into S.

Solution:

- **(a) Proof (Reducing / Minus Theorem):** Since S spans V and is not a basis, S must be linearly dependent. By the characterization theorem (Q4.6(b)), some vector $\mathbf{v} \in S$ is a linear combination of the other vectors in S. Removing $\mathbf{v}$ yields a proper subset $S_1 = S \setminus \{\mathbf{v}\}$ that still spans V (any linear combination involving $\mathbf{v}$ can replace $\mathbf{v}$ by its expression in S_1).
  - If S_1 is linearly independent, then $B = S_1 \subset S$ is a basis for V and we stop.
  - Otherwise, repeat this removal step. Since S is finite and $V \neq \{\mathbf{0}\}$ (so a spanning set cannot become empty), this process must terminate after finitely many steps at a linearly independent subset $B \subset S$ that still spans V—a basis for V. ■
- **(b) Proof (Extending / Plus Theorem):** Let $\dim(V) = n$. Since S is linearly independent and not already a basis for V, S does not span V. Thus there exists some vector $\mathbf{w}_1 \in V$ such that $\mathbf{w}_1 \notin \text{span}(S)$. Let $S_1 = S \cup \{\mathbf{w}_1\}$.
  - Claim: S_1 is linearly independent (if $c_1\mathbf{v}_1 + \cdots + c_r\mathbf{v}_r + d\mathbf{w}_1 = \mathbf{0}$, then d must be 0, else $\mathbf{w}_1 \in \text{span}(S)$; and $d = 0 \implies c_1 = \cdots = c_r = 0$ because S is LI).
  - If S_1 spans V, then S_1 is a basis and we stop. Otherwise, repeat by picking $\mathbf{w}_2 \notin \text{span}(S_1)$.
  - Because V is n-dimensional, no linearly independent set in V can contain more than n vectors (Theorem Q4.8(a)). Hence this process must terminate after at most $n - |S|$ insertions at a linearly independent set that spans V—a basis for V. ■

Q4.10 [PROOF] The Half-Test for Bases & Subspace Dimension Theorem

Sources:

- *Spring 2022 FE Q5(a),(b) [5+5 Marks]*
- *Spring 2024 FE Q4 [10 Marks]*

> - **(a) (Half-Test / Sp22 FE Q5a, Sp24 FE Q4):** Prove: If V is an n-dimensional vector space and S is a set in V with exactly n vectors, then S is a basis for V if **either** S spans V **or** S is linearly independent.
> - **(b) (Sp22 FE Q5b):** Prove: If W is a subspace of a finite-dimensional vector space V, then $\dim(W) \leq \dim(V)$; moreover, if $\dim(W) = \dim(V)$, then $W = V$.

Solution:

- **(a) Proof of Half-Test:** Let $|S| = n$ in an n-dimensional space V.
  - *Suppose S spans V:* If S were linearly dependent, by the Minus Theorem (Q4.9a) we could remove at least one vector from S to obtain a basis of V having $< n$ vectors, contradicting $\dim(V) = n$ (every basis of V has size n). Hence S must be linearly independent, so S is a basis.
  - *Suppose S is linearly independent:* If S did not span V, we could find $\mathbf{v} \notin \text{span}(S)$ and form a linearly independent set $S \cup \{\mathbf{v}\}$ with $n + 1$ vectors, contradicting Q4.8(a) (no LI set in V has $> n$ vectors). Hence S must span V, so S is a basis. ■
- **(b) Proof of Subspace Dimension Theorem:** Let $\dim(V) = n$.
  - If $W = \{\mathbf{0}\}$, then $\dim(W) = 0 \leq n$.
  - If $W \neq \{\mathbf{0}\}$, pick any nonzero $\mathbf{v}_1 \in W$ (linearly independent). Either $\{\mathbf{v}_1\}$ spans W, or we can successively add $\mathbf{v}_k \in W \setminus \text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}\}$, keeping the set linearly independent inside $W \subseteq V$. Since V has no linearly independent set of size $> n$, this process must stop at some basis B of W with $|B| \leq n$, proving $\dim(W) \leq \dim(V)$.
  - If $\dim(W) = \dim(V) = n$, any basis B of W is a linearly independent set of n vectors inside V. By the Half-Test (part (a)), B is also a basis for V, so $W = \text{span}(B) = V$. ■

Q4.11 [CALC] Checking Bases ($\mathbb{R}^3$, P_2, Trig Space) & Extending to Bases

Sources: *Quiz 8 P1–P8; Quiz 9 P1–P3; Spring 2024 FE Q9(b) [4 Marks]; Spring 2023 FE Q5(b) [6 Marks]*

> **(A) $\mathbb{R}^3$ Basis Checks (Quiz 8 P3, P4, P7, P8):** Determine whether each set of 3 vectors is a basis for $\mathbb{R}^3$:
>
> 1. *(P3)* $\{(1,6,4),\ (2,4,-1),\ (-1,2,5)\}$
> 2. *(P4)* $\{(1,0,1),\ (-2,1,4),\ (-1,1,5)\}$
> 3. *(P7)* $\{(1,0,1),\ (-2,1,4),\ (0,3,1)\}$
> 4. *(P8)* $\{(2,3,1),\ (4,1,1),\ (0,-7,1)\}$
>
> **(B) P_2 Basis Checks (Quiz 8 P5, P6; Sp24 FE Q9b):** Determine whether each set is a basis for P_2: 5. *(Quiz 8 P5)* $\{1-3x+2x^2,\ 1+x+4x^2,\ 1-7x\}$ 6. *(Quiz 8 P6 & Sp24 FE Q9b)* $\{4+6x+x^2,\ -1+4x+2x^2,\ 5+2x-x^2\}$
>
> **(C) Trigonometric Vector Space (Sp23 FE Q5b; Quiz 8 P1, P2):** 7. Let V be spanned by $S=\{\mathbf{v}_1,\mathbf{v}_2,\mathbf{v}_3\}=\{\cos^2 x,\ \sin^2 x,\ \cos 2x\}$.
>
> - (i) Show that S is not a basis for V.
> - (ii) Find a basis for V (Quiz 8 P1 asks for one containing $\cos 2x$; Quiz 8 P2 asks for one containing $\sin^2 x$).
>
> **(D) Prove Linear Independence & Extend to a Basis for $\mathbb{R}^3$ (Quiz 9 P1–P3):** 8. *(P1)* $S=\{(1,0,1)^T,\ (2,1,3)^T\}$ 9. *(P2)* $S=\{(1,2,0)^T,\ (3,5,1)^T\}$ 10. *(P3)* $S=\{(2,1,-1)^T,\ (-1,3,4)^T\}$

Solutions: (A) $\mathbb{R}^3$ Basis Checks (by Half-Test: 3 vectors form a basis iff $\det(A)\neq 0$):

1. $\det\begin{bmatrix}1&2&-1\\6&4&2\\4&-1&5\end{bmatrix}=1(22)-2(22)-1(-22)=0 \implies$ **No** (note $\mathbf{v}_3=\mathbf{v}_1-\mathbf{v}_2$).
2. $\det\begin{bmatrix}1&-2&-1\\0&1&1\\1&4&5\end{bmatrix}=1(5-4)+2(0-1)-1(0-1)=1-2+1=0 \implies$ **No** (note $\mathbf{v}_3=\mathbf{v}_1+\mathbf{v}_2$; Quiz 8 key had `-(0+1)` typo inside parentheses, result 0 is right).
3. $\det\begin{bmatrix}1&-2&0\\0&1&3\\1&4&1\end{bmatrix}=1(1-12)+2(0-3)=-17\neq 0 \implies$ **Yes**.
4. $\det\begin{bmatrix}2&4&0\\3&1&-7\\1&1&1\end{bmatrix}=2(1+7)-4(3+7)=-24\neq 0 \implies$ **Yes**.

(B) P_2 Basis Checks ($\dim(P_2)=3$, check coordinate determinant in $\{1,x,x^2\}$): 5. Coordinate matrix $A=\begin{bmatrix}1&1&1\\-3&1&-7\\2&4&0\end{bmatrix}$, $\det(A)=1(28)-1(14)+1(-14)=0 \implies$ **No** (not a basis; $2p_1-p_2=p_3$). 6. Coordinate matrix $A=\begin{bmatrix}4&-1&5\\6&4&2\\1&2&-1\end{bmatrix}$, $\det(A)=4(-8)+1(-8)+5(8)=0 \implies$ **No** (not a basis; $p_1-p_2=p_3$).

(C) Trigonometric Space: 7. **(i)** Using the identity $\cos 2x=\cos^2 x-\sin^2 x$, we have the nontrivial relation:

$$1\cdot\cos^2 x+(-1)\cdot\sin^2 x+(-1)\cdot\cos 2x=0.$$

Hence S is **linearly dependent** and cannot be a basis for V. **(ii)** Because $\cos 2x$ (or $\cos^2 x$) is a linear combination of the other two, removing any one redundant vector preserves the span V (by the Minus Theorem):

- *Basis containing* $\cos^2 x,\sin^2 x$ *(Sp23 FE Q5b):* $\{\cos^2 x,\sin^2 x\}$. Linear independence: $c_1\cos^2 x+c_2\sin^2 x=0 \implies$ at $x=0$: $c_1=0$; at $x=\frac{\pi}{2}$: $c_2=0$.
- *Basis containing* $\cos 2x$ *(Quiz 8 P1):* $\{\cos^2 x,\cos 2x\}$ or $\{\sin^2 x,\cos 2x\}$. Check LI of $\{\sin^2 x,\cos 2x\}$: $a\sin^2 x+b\cos 2x=0 \implies$ at $x=0$: $b=0$; at $x=\frac{\pi}{2}$: $a=0$. (Similarly $\cos^2 x$ and $\cos 2x$ are not scalar multiples.)

(D) Prove LI & Extend to $\mathbb{R}^3$: 8. *(P1)* $a(1,0,1)^T+b(2,1,3)^T=\mathbf{0} \implies b=0$ (from 2nd row), then $a=0$ (from 1st row) $\implies$ LI. Add $\mathbf{w}=(x,y,z)^T$: $\det\begin{bmatrix}1&2&x\\0&1&y\\1&3&z\end{bmatrix}=-x-y+z$. Choosing $\mathbf{w}=(0,0,1)^T$ gives $\det=1\neq 0 \implies$ Basis: $\{(1,0,1)^T,\ (2,1,3)^T,\ (0,0,1)^T\}$. 9. *(P2)* $a(1,2,0)^T+b(3,5,1)^T=\mathbf{0} \implies b=0$ (3rd row), then $a=0 \implies$ LI. Add $\mathbf{w}=(x,y,z)^T$: $\det\begin{bmatrix}1&3&x\\2&5&y\\0&1&z\end{bmatrix}=2x-y-z$. Choosing $\mathbf{w}=(0,0,1)^T$ gives $\det=-1\neq 0 \implies$ Basis: $\{(1,2,0)^T,\ (3,5,1)^T,\ (0,0,1)^T\}$. *(Note: Quiz 9 key forgot the minus sign in* $\det$*, writing* $+1$*, though* $-1\neq 0$ *still makes* $(0,0,1)^T$

valid!) 10. *(P3)* $a(2,1,-1)^T + b(-1,3,4)^T = \mathbf{0} \implies 2a - b = 0$ and $a + 3b = 0 \implies a = b = 0 \implies$ LI. Add $\mathbf{w} = (x,y,z)^T$: $\det \begin{bmatrix} 2 & -1 & x \\ 1 & 3 & y \\ -1 & 4 & z \end{bmatrix} = 7x - 7y + 7z$. Choosing $\mathbf{w} = (0,0,1)^T$ gives $\det = 7 \neq 0 \implies$ Basis: $\{(2,1,-1)^T,\ (-1,3,4)^T,\ (0,0,1)^T\}$.

Part 5: Row Space, Column Space, Null Space, Rank–Nullity & Consistency

Q5.1 [PROOF] Structure of General Solution to $A\mathbf{x} = \mathbf{b}$

Source: *Spring 2022 Midterm Q4 [10 Marks]*

> Let $A\mathbf{x} = \mathbf{b}$ be any consistent system of linear equations and let $\mathbf{x}_1$ be a fixed solution. Show that every solution to the system can be written in the form $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_0$, where $\mathbf{x}_0$ is a solution of the homogeneous system $A\mathbf{x} = \mathbf{0}$. Show also that every vector of this form is a solution.

Solution:

1. **Every solution has the form $\mathbf{x}_1 + \mathbf{x}_0$:** Let $\mathbf{x}$ be any solution to $A\mathbf{x} = \mathbf{b}$. Define $\mathbf{x}_0 = \mathbf{x} - \mathbf{x}_1$. Then:
$$A\mathbf{x}_0 = A(\mathbf{x} - \mathbf{x}_1) = A\mathbf{x} - A\mathbf{x}_1 = \mathbf{b} - \mathbf{b} = \mathbf{0},$$
so $\mathbf{x}_0$ is a solution of $A\mathbf{x} = \mathbf{0}$ (i.e., $\mathbf{x}_0 \in \text{Null}(A)$), and $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_0$.
2. **Every vector $\mathbf{x}_1 + \mathbf{x}_0$ is a solution:** Suppose $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_0$ where $A\mathbf{x}_1 = \mathbf{b}$ and $A\mathbf{x}_0 = \mathbf{0}$. Then:
$$A\mathbf{x} = A(\mathbf{x}_1 + \mathbf{x}_0) = A\mathbf{x}_1 + A\mathbf{x}_0 = \mathbf{b} + \mathbf{0} = \mathbf{b}. \quad \blacksquare$$

Q5.2 [PROOF] Consistency Criterion via Column Space

Sources: *Spring 2023 FE Q2(a) [7 Marks]; Spring 2025 FE Solution P4(b)*

> Prove that a system of linear equations $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is in the column space of A.

Solution: Write $A = [\mathbf{c}_1 \ \ \mathbf{c}_2 \ \ \cdots \ \ \mathbf{c}_n]$ in terms of its columns $\mathbf{c}_1, \ldots, \mathbf{c}_n$, and $\mathbf{x} = [x_1, x_2, \ldots, x_n]^T$. By the column-combination view of matrix multiplication:

$$A\mathbf{x} = x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + \cdots + x_n\mathbf{c}_n.$$

Thus:

- **($\Rightarrow$)** If $A\mathbf{x} = \mathbf{b}$ is consistent, there exists some vector $\mathbf{x} = [x_1, \ldots, x_n]^T$ such that $\mathbf{b} = x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n$, which is a linear combination of the columns of A; hence $\mathbf{b} \in \text{Col}(A)$.
- **($\Leftarrow$)** If $\mathbf{b} \in \text{Col}(A)$, then by definition of $\text{Col}(A) = \text{span}\{\mathbf{c}_1, \ldots, \mathbf{c}_n\}$, there exist scalars $x_1, \ldots, x_n$ such that $\mathbf{b} = x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n = A\mathbf{x}$. Hence $\mathbf{x} = [x_1, \ldots, x_n]^T$ is a solution, so $A\mathbf{x} = \mathbf{b}$ is consistent. $\blacksquare$

Q5.3 [PROOF] Consistency for All $\mathbf{b} \iff A$ Invertible & Master Theorem Cycle

Sources: *Spring 2022 FE Q8(a),(b) [7+3 Marks]; Spring 2024 Mid Part A Q2 [10 Marks]; Spring 2024 FE Q1 [12 Marks]; Spring 2023 Mid Q7 [10 Marks]*

> - **(A) (Sp22 FE Q8; Sp24 Mid A Q2):** Let A be an $n \times n$ matrix. Prove that $A\mathbf{x} = \mathbf{b}$ is consistent for **every** $n \times 1$ vector $\mathbf{b}$ if and only if A is invertible (non-singular).
> - **(B) (Sp24 FE Q1; Sp23 Mid Q7):** Prove the Master Theorem cycle $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (a)$ for an $n \times n$ matrix A :
>   - *(a)* $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
>   - *(b)* $\text{RREF}(A) = I_n$.
>   - *(c)* A is expressible as a product of elementary matrices.
>   - *(d)* A is invertible (non-singular).

Solution: (A) Proof of Consistency for all $\mathbf{b} \iff A$ Invertible:

- **(⇒) (Sp22 FE Q8a / Sp24 Mid A Q2):** Suppose $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^n$. In particular, for each standard basis vector $\mathbf{e}_j$ ($j = 1, \ldots, n$), there exists a solution $\mathbf{c}_j \in \mathbb{R}^n$ such that $A\mathbf{c}_j = \mathbf{e}_j$. Form the $n \times n$ matrix $C = [\mathbf{c}_1 \;\; \mathbf{c}_2 \;\; \cdots \;\; \mathbf{c}_n]$. Then:

  $$AC = [A\mathbf{c}_1 \;\; A\mathbf{c}_2 \;\; \cdots \;\; A\mathbf{c}_n] = [\mathbf{e}_1 \;\; \mathbf{e}_2 \;\; \cdots \;\; \mathbf{e}_n] = I_n.$$

  Hence $\det(A)\det(C) = \det(I_n) = 1 \neq 0$, so $\det(A) \neq 0$ and A is invertible (with $A^{-1} = C$). *(Rank/Col-space view: consistency for all $\mathbf{b}$ means $\mathrm{Col}(A) = \mathbb{R}^n$, so $\mathrm{rank}(A) = n$, hence $\mathrm{nullity}(A) = 0$ and A is invertible.)*
- **(⇐) (Sp22 FE Q8b):** If A is invertible, then for any $\mathbf{b} \in \mathbb{R}^n$, take $\mathbf{x} = A^{-1}\mathbf{b}$. Then $A\mathbf{x} = A(A^{-1}\mathbf{b}) = I_n\mathbf{b} = \mathbf{b}$, so $A\mathbf{x} = \mathbf{b}$ is consistent. ■

(B) Proof of Master Theorem Cycle:

- $(a) \Rightarrow (b)$**:** If $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$, then $\mathrm{RREF}(A)$ has no free variables, meaning every one of the n columns is a pivot column. Since $\mathrm{RREF}(A)$ is $n \times n$ with n leading 1's (one in every row and column, and 0's above/below pivots), $\mathrm{RREF}(A) = I_n$.
- $(b) \Rightarrow (c)$**:** Reducing A to $\mathrm{RREF}(A) = I_n$ applies a finite sequence of elementary row operations corresponding to elementary matrices $E_1, \ldots, E_k$, so $E_k \cdots E_2 E_1 A = I_n$. Since each elementary matrix is invertible and its inverse is also elementary, $A = E_1^{-1}E_2^{-1} \cdots E_k^{-1}$ is a product of elementary matrices.
- $(c) \Rightarrow (d)$**:** Each elementary matrix is invertible, and a finite product of invertible matrices is invertible (with $(E_1^{-1} \cdots E_k^{-1})^{-1} = E_k \cdots E_1$). Thus A is invertible.
- $(d) \Rightarrow (a)$**:** If A is invertible and $A\mathbf{x} = \mathbf{0}$, multiplying on the left by A^{-1} gives $\mathbf{x} = A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{0} = \mathbf{0}$. So only the trivial solution exists. ■

Q5.4 [PROOF] Dimension of Row Space Equals Dimension of Column Space & $\mathrm{rank}(A) = \mathrm{rank}(A^T)$

Source: *Spring 2023 FE Q1(a),(b) [7+3 Marks]*

> - **(a)** Prove: If A is any matrix, then the row space and column space of A have the same dimension ($\dim(\mathrm{Row}(A)) = \dim(\mathrm{Col}(A))$).
> - **(b)** For a matrix A (asked for square A in the exam), show that A and A^T have the same rank.

Solution:

- **(a)** Let $R = \mathrm{RREF}(A)$, obtained from A via elementary row operations, and let r be the number of nonzero rows (pivots) in R.

  1. **Row space dimension:** Elementary row operations do not change the row space of a matrix, so $\mathrm{Row}(A) = \mathrm{Row}(R)$. The r nonzero rows of R are linearly independent (each pivot 1 sits in a column where all other rows have 0) and span $\mathrm{Row}(R)$. Hence $\dim(\mathrm{Row}(A)) = r$.
  2. **Column space dimension:** Elementary row operations preserve the solution set of the homogeneous system $A\mathbf{x} = \mathbf{0} \iff R\mathbf{x} = \mathbf{0}$. Equivalently, any linear dependence relation among the columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ of A,

     $$x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0},$$

     holds **if and only if** the exact same relation holds among the columns $\mathbf{r}_1, \ldots, \mathbf{r}_n$ of R:

     $$x_1\mathbf{r}_1 + \cdots + x_n\mathbf{r}_n = \mathbf{0}.$$

     In R, the r pivot columns are the standard basis vectors $\mathbf{e}_1, \ldots, \mathbf{e}_r$ (with trailing 0's), which are linearly independent and span all columns of R. Therefore, the corresponding r pivot columns of A are also linearly independent and every other column of A is a linear combination of them—forming a basis of $\mathrm{Col}(A)$. Hence $\dim(\mathrm{Col}(A)) = r$.
  3. Combining (1) and (2), $\dim(\mathrm{Row}(A)) = r = \dim(\mathrm{Col}(A))$. ■

- **(b)** Notice the rows of A^T are the columns of A, and the columns of A^T are the rows of A, so $\mathrm{Col}(A^T) = \mathrm{Row}(A)$. Using part (a):

  $$\mathrm{rank}(A^T) = \dim(\mathrm{Col}(A^T)) = \dim(\mathrm{Row}(A)) = \dim(\mathrm{Col}(A)) = \mathrm{rank}(A). \quad ■$$

Q5.5 [CALC] Basis for $\mathrm{Row}(A)$ Consisting Entirely of Row Vectors of A

Source: *Spring 2023 FE Q9 [10 Marks]*

For the matrix $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -3 & -2 & 0 \\ 10 & -5 & 15 & 10 & 0 \\ 22 & -10 & 18 & 8 & 0 \end{bmatrix}$, find a basis for the row space of A consisting **entirely of row vectors from** A.

Solution: Recall that finding a basis of $\mathrm{Row}(A)$ out of original rows of A is equivalent to finding a basis of $\mathrm{Col}(A^T)$ out of columns of A^T! Row-reduce A^T:

$$A^T = \begin{bmatrix} 1 & 0 & 10 & 22 \\ 0 & 1 & -5 & -10 \\ 0 & -3 & 15 & 18 \\ 0 & -2 & 10 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_2]{R_3 \leftarrow R_3 + 3R_2} \begin{bmatrix} 1 & 0 & 10 & 22 \\ 0 & 1 & -5 & -10 \\ 0 & 0 & 0 & -12 \\ 0 & 0 & 0 & -12 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \mathbf{1} & 0 & 10 & 0 \\ 0 & \mathbf{1} & -5 & 0 \\ 0 & 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns of A^T are **Columns 1, 2, and 4** (notice Column 3 $= 10\,\mathrm{Col}_1 - 5\,\mathrm{Col}_2$, i.e., $\mathbf{r}_3 = 10\mathbf{r}_1 - 5\mathbf{r}_2$ in A). Therefore, a basis for $\mathrm{Row}(A)$ consisting entirely of original rows of A is $\{\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_4\}$:

$$\text{Basis for } \mathrm{Row}(A) = \big\{(1,0,0,0,0),\ \ (0,1,-3,-2,0),\ \ (22,-10,18,8,0)\big\}.$$

Q5.6 [CALC] Null Space Bases, Nullity & Rank

Source: *Quiz 10 Problems 3–6 (+ Official Sol; missing computations for P4–P6 filled in & verified)*

Find a basis for the null space of each matrix A, and state its nullity / rank:

1. *(P3)* $A = \begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ 7 & -6 & 2 \end{bmatrix}$ *(find basis for $\mathrm{Null}(A)$ & nullity)*
2. *(P4)* $A = \begin{bmatrix} 2 & -3 & 1 \\ 4 & -5 & -2 \\ 6 & -9 & -1 \end{bmatrix}$ *(find basis for $\mathrm{Null}(A)$ & nullity)*
3. *(P5)* $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -2 & 6 \\ 3 & 6 & -3 & 9 \end{bmatrix}$ *(find basis for $\mathrm{Null}(A)$ & rank)*
4. *(P6)* $A = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 3 & 2 \\ 0 & 3 & 9 & 6 \end{bmatrix}$ *(find basis for $\mathrm{Null}(A)$ & rank)*

Solutions:

1. **(P3):** $A \xrightarrow[R_3 \leftarrow R_3 - 7R_1]{R_2 \leftarrow R_2 - 5R_1} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -19 \\ 0 & 1 & -19 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & -16 \\ 0 & 1 & -19 \\ 0 & 0 & 0 \end{bmatrix}$. Solve $A\mathbf{x} = \mathbf{0}$: $x_3 = t$ (free), $x_2 = 19t$, $x_1 = 16t$.
   - **Basis for $\mathrm{Null}(A)$:** $\left\{ \begin{bmatrix} 16 \\ 19 \\ 1 \end{bmatrix} \right\}$, $\mathrm{nullity}(A) = 1$.
2. **(P4)** *(Note: Official PDF wrote "Same as problem 3" for the method, **not** the answer — solving it gives a trivial null space!)*:

$$A = \begin{bmatrix} 2 & -3 & 1 \\ 4 & -5 & -2 \\ 6 & -9 & -1 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 - 3R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 2 & -3 & 1 \\ 0 & 1 & -4 \\ 0 & 0 & -4 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

   Every column is a pivot column ($\mathrm{rank}(A) = 3$), so $A\mathbf{x} = \mathbf{0}$ has only $\mathbf{x} = \mathbf{0}$.
   - $\mathrm{Null}(A) = \{\mathbf{0}\}$, **basis:** $\emptyset$ (the empty set; $\{\mathbf{0}\}$ is 0-dimensional), $\mathrm{nullity}(A) = 0$.
3. **(P5):** Rows 2 and 3 are $2R_1$ and $3R_1$, so:

$$\mathrm{RREF}(A) = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

   Free variables: $x_2 = r,\ x_3 = s,\ x_4 = t$, with $x_1 = -2r + s - 3t$:

$$\mathbf{x} = r\begin{bmatrix}-2\\1\\0\\0\end{bmatrix} + s\begin{bmatrix}1\\0\\1\\0\end{bmatrix} + t\begin{bmatrix}-3\\0\\0\\1\end{bmatrix}.$$

- **Basis for** $\mathrm{Null}(A)$: $\left\{\begin{bmatrix}-2\\1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\\0\end{bmatrix}, \begin{bmatrix}-3\\0\\0\\1\end{bmatrix}\right\}$, $\mathrm{rank}(A) = 1$ ($\mathrm{nullity}(A) = 3$).

4. **(P6):** Row 3 is $3R_2$, so:

$$\mathrm{RREF}(A) = \begin{bmatrix}1 & 0 & 2 & -1\\0 & 1 & 3 & 2\\0 & 0 & 0 & 0\end{bmatrix}.$$

Free variables: $x_3 = s,\ x_4 = t$, with $x_1 = -2s + t$, $x_2 = -3s - 2t$:

$$\mathbf{x} = s\begin{bmatrix}-2\\-3\\1\\0\end{bmatrix} + t\begin{bmatrix}1\\-2\\0\\1\end{bmatrix}.$$

- **Basis for** $\mathrm{Null}(A)$: $\left\{\begin{bmatrix}-2\\-3\\1\\0\end{bmatrix}, \begin{bmatrix}1\\-2\\0\\1\end{bmatrix}\right\}$, $\mathrm{rank}(A) = 2$ ($\mathrm{nullity}(A) = 2$).

Q5.7 [CALC] Rank–Nullity Dimension Table Problems

Sources: *Spring 2024 FE Q9(a) [6 Marks]; Quiz 10 Problems 1, 2, 7, 8 (+ Official Sol)*

> Use the Rank–Nullity Theorem ($\mathrm{rank}(A) + \mathrm{nullity}(A) = n$ for an $m \times n$ matrix, and $\mathrm{rank}(A^T) = \mathrm{rank}(A)$) to find the requested dimensions:
>
> - **Case 1 (Sp24 FE Q9a(i) & Quiz 10 P1):** A is 5×9 with $\mathrm{rank}(A) = 2$. Find $\dim \mathrm{Row}(A)$, $\dim \mathrm{Col}(A)$, $\dim \mathrm{Null}(A)$, and $\dim \mathrm{Null}(A^T)$.
> - **Case 2 (Sp24 FE Q9a(ii) & Quiz 10 P8):** A is 6×2 with $\mathrm{rank}(A) = 2$. Find $\dim \mathrm{Row}(A)$, $\dim \mathrm{Col}(A)$, $\dim \mathrm{Null}(A)$ (Sp24 FE), and for A^T (2×6) find $\dim \mathrm{Row}(A^T), \dim \mathrm{Col}(A^T), \dim \mathrm{Null}(A^T)$ (Quiz 10 P8).
> - **Case 3 (Quiz 10 P2):** A is 6×8 with $\mathrm{rank}(A) = 3$. Find $\dim \mathrm{Row}(A)$, $\dim \mathrm{Col}(A)$, $\dim \mathrm{Null}(A)$, and $\dim \mathrm{Null}(A^T)$.
> - **Case 4 (Quiz 10 P7):** A is 9×5 with $\mathrm{rank}(A) = 2$. Find $\dim \mathrm{Row}(A^T)$, $\dim \mathrm{Col}(A^T)$, $\dim \mathrm{Null}(A^T)$.

Solutions:

Case	Size of A	$\mathrm{rank}(A)$	$\dim \mathrm{Row}(A)$	$\dim \mathrm{Col}(A)$	$\dim \mathrm{Null}(A) = n - r$	$\dim \mathrm{Row}(A^T) = \dim \mathrm{Col}(A^T)$	$\dim \mathrm{Null}(A^T) = m - r$
Case 1	5×9	2	2	2	$9 - 2 = \mathbf{7}$	2	$5 - 2 = \mathbf{3}$
Case 2	6×2	2	2	2	$2 - 2 = \mathbf{0}$	2	$6 - 2 = \mathbf{4}$
Case 3	6×8	3	3	3	$8 - 3 = \mathbf{5}$	3	$6 - 3 = \mathbf{3}$
Case 4	9×5	2	2	2	$5 - 2 = \mathbf{3}$	2	$9 - 2 = \mathbf{7}$

Part 6: Linear Transformations, Standard Matrices, Kernel & Range

Q6.1 [PROOF / T-F] Linear Transformation Theorems & Counterexamples

Source: *Spring 2023 FE Q6(c),(d),(e) [2 each], Q2(b) [3 Marks]*

> Evaluate and prove/justify each statement:
>
> - **(a) (Sp23 FE Q6c):** Let $T : M_{nn} \to \mathbb{R}$ be given by $T(A) = \det(A)$. True or False: T is a linear transformation.
> - **(b) (Sp23 FE Q6d):** If $T : V \to W$ is a linear transformation, prove T is one-to-one if and only if $\ker(T) = \{\mathbf{0}\}$.
> - **(c) (Sp23 FE Q6e):** Every possible linear transformation between Euclidean spaces ($\mathbb{R}^n \to \mathbb{R}^m$) can be represented by some matrix.
> - **(d) (Sp23 FE Q2b):** Using Q5.2 ($A\mathbf{x} = \mathbf{b}$ consistent $\iff$ $\mathbf{b} \in \text{Col}(A)$), explain why for $T_A(\mathbf{x}) = A\mathbf{x}$, $\dim(R(T_A)) = \text{rank}(A)$.

Solution:

- **(a) FALSE (for $n \geq 2$).** Let $A = B = I_2 \in M_{2\times 2}$. Then:

$$T(A+B) = \det(2I_2) = 4, \qquad T(A) + T(B) = \det(I_2) + \det(I_2) = 1 + 1 = 2 \neq 4.$$

  Since $T(A+B) \neq T(A) + T(B)$, T fails additivity (and also fails homogeneity: $\det(cA) = c^n \det(A) \neq c\det(A)$ for $n \geq 2$). *(Edge case note: for $n = 1$, $T([a]) = a$ IS linear; for general $n \geq 2$ the statement is False.)*

- **(b) ALWAYS TRUE — Proof:**
  - *($\Rightarrow$)* Suppose T is one-to-one. First, $T(\mathbf{0}) = \mathbf{0}$ (by $T(0\mathbf{v}) = 0T(\mathbf{v}) = \mathbf{0}$), so $\mathbf{0} \in \ker(T)$. If $\mathbf{v} \in \ker(T)$, then $T(\mathbf{v}) = \mathbf{0} = T(\mathbf{0})$. Since T is one-to-one, $\mathbf{v} = \mathbf{0}$. Hence $\ker(T) = \{\mathbf{0}\}$.
  - *($\Leftarrow$)* Suppose $\ker(T) = \{\mathbf{0}\}$, and assume $T(\mathbf{u}) = T(\mathbf{v})$ for some $\mathbf{u}, \mathbf{v} \in V$. By linearity:

$$T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0} \implies \mathbf{u} - \mathbf{v} \in \ker(T) = \{\mathbf{0}\} \implies \mathbf{u} - \mathbf{v} = \mathbf{0} \implies \mathbf{u} = \mathbf{v}.$$

    Thus T is one-to-one. ■

- **(c) ALWAYS TRUE — Proof:** Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear, and let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be the standard basis of $\mathbb{R}^n$. Every $\mathbf{x} = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$ equals $x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$. By linearity:

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) + \cdots + x_nT(\mathbf{e}_n) = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{bmatrix}\mathbf{x} = [T]\mathbf{x}.$$

  So every linear map $\mathbb{R}^n \to \mathbb{R}^m$ is matrix multiplication by its $m \times n$ standard matrix $[T]$. ■

- **(d) Explanation:** The range of T_A is $R(T_A) = \{\mathbf{b} \in \mathbb{R}^m : \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\}$, which is precisely the set of $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ is consistent. By Q5.2, $A\mathbf{x} = \mathbf{b}$ is consistent $\iff$ $\mathbf{b} \in \text{Col}(A)$. Hence $R(T_A) = \text{Col}(A)$, so:

$$\dim(R(T_A)) = \dim(\text{Col}(A)) = \text{rank}(A). \quad ■$$

Q6.2 [CALC] Finding the Standard Matrix $[T]$ from Standard Basis Images

Sources: *Spring 2022 FE Q9(a)–(c) [10 Marks]; Spring 2025 FE Solution P7(b)*

> Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation with:
>
> - **Version A (Sp22 FE Q9):** $T(\mathbf{e}_1) = (1, 1),\ T(\mathbf{e}_2) = (3, 0),\ T(\mathbf{e}_3) = (4, 7)$. Find: (a) $T(2, 4, 6)$, (b) $T(x, y, z)$, (c) the matrix of T.
> - **Version B (Sp25 FE P7b):** $T(\mathbf{e}_1) = (1, 1),\ T(\mathbf{e}_2) = (3, 0),\ T(\mathbf{e}_3) = (4, -7)$. Find $[T]$.

Solution:

- **Version A:**
  - *(a)* $T(2, 4, 6) = 2T(\mathbf{e}_1) + 4T(\mathbf{e}_2) + 6T(\mathbf{e}_3) = 2(1, 1) + 4(3, 0) + 6(4, 7) = (2 + 12 + 24,\ 2 + 0 + 42) = (\mathbf{38}, \mathbf{44})$.
  - *(b)* $T(x, y, z) = x(1, 1) + y(3, 0) + z(4, 7) = (\mathbf{x + 3y + 4z},\ \mathbf{x + 7z})$.
  - *(c)* $[T] = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & T(\mathbf{e}_3) \end{bmatrix} = \begin{bmatrix} 1 & 3 & 4 \\ 1 & 0 & 7 \end{bmatrix}$.
- **Version B (Sp25 FE P7b):**
  - $[T] = \begin{bmatrix} 1 & 3 & 4 \\ 1 & 0 & -7 \end{bmatrix}$.

Q6.3 [CALC] Formula and Matrix of T from a Custom Basis

Source: *Fall 2022 FE Q6(a)–(c) [5+2+3 Marks]*

> Consider the basis $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for $\mathbb{R}^3$, where $\mathbf{v}_1 = (1,1,1)$, $\mathbf{v}_2 = (1,1,0)$, and $\mathbf{v}_3 = (1,0,0)$. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be the linear transformation such that:
>
> $$T(\mathbf{v}_1) = (1,0), \qquad T(\mathbf{v}_2) = (2,-1), \qquad T(\mathbf{v}_3) = (4,3).$$
>
> - **(a)** Find a formula for $T(x_1, x_2, x_3)$.
> - **(b)** Use your formula from part (a) to find $T(1,-5,0)$.
> - **(c)** Find the matrix of T.

Solution:

- **(a)** Write an arbitrary $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$ as $\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$:

  $$(x_1, x_2, x_3) = c_1(1,1,1) + c_2(1,1,0) + c_3(1,0,0) = (c_1 + c_2 + c_3,\ c_1 + c_2,\ c_1).$$

  Equating components gives back-substitution immediately:

  - $c_1 = x_3$
  - $c_2 = x_2 - c_1 = x_2 - x_3$
  - $c_3 = x_1 - (c_1 + c_2) = x_1 - x_2$.

  Now apply linearity of T:

  $$\begin{aligned} T(x_1, x_2, x_3) &= c_1 T(\mathbf{v}_1) + c_2 T(\mathbf{v}_2) + c_3 T(\mathbf{v}_3) \\ &= x_3(1,0) + (x_2 - x_3)(2,-1) + (x_1 - x_2)(4,3) \\ &= \big(x_3 + 2x_2 - 2x_3 + 4x_1 - 4x_2,\ \ 0 - x_2 + x_3 + 3x_1 - 3x_2\big) \\ &= (\mathbf{4x_1 - 2x_2 - x_3},\ \ \mathbf{3x_1 - 4x_2 + x_3}). \end{aligned}$$

  (Sanity check: $T(1,1,1) = (4-2-1, 3-4+1) = (1,0)$; $T(1,1,0) = (2,-1)$; $T(1,0,0) = (4,3)$. ✓)

- **(b)** $T(1,-5,0) = \big(4(1) - 2(-5) - 0,\ \ 3(1) - 4(-5) + 0\big) = (\mathbf{14}, \mathbf{23})$.

- **(c)** Reading the coefficients of x_1, x_2, x_3 (or evaluating $T(\mathbf{e}_1), T(\mathbf{e}_2), T(\mathbf{e}_3)$):

  $$[T] = \begin{bmatrix} 4 & -2 & -1 \\ 3 & -4 & 1 \end{bmatrix}.$$

Q6.4 [CALC] Finding $\ker(T_A)$ for a Matrix Transformation

Source: *Spring 2025 FE Solution Problem 5*

> Let $T_A : \mathbb{R}^4 \to \mathbb{R}^3$ be the linear transformation defined by $A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$. Calculate $\ker(T_A)$.

Solution (Official Key): Row-reduce the augmented matrix $[A \mid \mathbf{0}]$ to RREF:

$$\begin{bmatrix} 1 & 4 & 5 & 2 & 0 \\ 2 & 1 & 3 & 0 & 0 \\ -1 & 3 & 2 & 2 & 0 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 4 & 5 & 2 & 0 \\ 0 & -7 & -7 & -4 & 0 \\ 0 & 7 & 7 & 4 & 0 \end{bmatrix} \xrightarrow[R_2 \leftarrow -\frac{1}{7}R_2]{R_3 \leftarrow R_3 + R_2} \begin{bmatrix} 1 & 4 & 5 & 2 & 0 \\ 0 & 1 & 1 & \frac{4}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 - 4R_2} \begin{bmatrix} 1 & 0 & 1 & -\frac{2}{7} & 0 \\ 0 & 1 & 1 & \frac{4}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Let $x_3 = s, x_4 = t$ (free variables). Then $x_1 = -s + \frac{2}{7}t$ and $x_2 = -s - \frac{4}{7}t$:

$$\ker(T_A) = \operatorname{span}\left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix},\ \begin{bmatrix} \frac{2}{7} \\ -\frac{4}{7} \\ 0 \\ 1 \end{bmatrix} \right\} \qquad (\dim(\ker T_A) = 2).$$

Q6.5 [CALC] Bases and Dimensions of $\ker(T)$ and $R(T)$

Source: *Spring 2024 FE Q6 [12 Marks]*

> Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the matrix transformation $T(\mathbf{x}) = A\mathbf{x}$ defined by:
>
> $$T(x_1, x_2, x_3) = (x_1 + x_2 + 2x_3,\ \ x_1 + x_3,\ \ 2x_1 + x_2 + 3x_3).$$
>
> Find bases for the kernel and range of T. Hence, state $\dim(\ker(T))$ and $\dim(R(T))$.

Solution: Extract the standard matrix A of T and row-reduce it to RREF:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{bmatrix} \xrightarrow[R_1 \leftarrow R_1 - R_2]{\substack{R_2 \leftarrow -R_2 \\ R_3 \leftarrow R_3 + R_2}} \begin{bmatrix} \mathbf{1} & 0 & 1 \\ 0 & \mathbf{1} & 1 \\ 0 & 0 & \mathbf{0} \end{bmatrix}.$$

- **Kernel ($\ker(T) = \text{Null}(A)$):** Solving $A\mathbf{x} = \mathbf{0}$ gives $x_1 = -x_3,\ x_2 = -x_3$ with $x_3 = t$ free:

$$\mathbf{x} = t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \implies \textbf{Basis for } \ker(T) = \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}, \quad \dim(\ker(\mathbf{T})) = \mathbf{1}.$$

- **Range ($R(T) = \text{Col}(A)$):** Pivot columns of $\text{RREF}(A)$ are Columns 1 and 2, so the first two columns of the **original** matrix A form a basis for $\text{Col}(A) = R(T)$:

$$\textbf{Basis for } R(T) = \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad \dim(\mathbf{R}(\mathbf{T})) = \mathbf{2}.$$

  (Check Dimension Theorem: $\dim(\ker(T)) + \dim(R(T)) = 1 + 2 = 3 = \dim(\mathbb{R}^3)$. ✓*)*

Q6.6 `[CALC / CONCEPT]` Polynomial Multiplication Transformation $T(p(x)) = x\,p(x)$

Source: *Fall 2022 FE Q10(a)–(c) [3+4+3 Marks]*

> Let $T : P_2 \to P_3$ be the linear transformation given by $T(p(x)) = x\,p(x)$.
>
> - **(a)** Which of x, 0, and $(1 - x)$ are in $\ker(T)$?
> - **(b)** Is T an onto transformation? Why or why not?
> - **(c)** What is the dimension of $\ker(T)$? Can you suggest a basis for it?

Solution:

- **(a)** A polynomial $p(x) \in P_2$ is in $\ker(T)$ iff $T(p(x)) = x\,p(x) = 0$ (the zero polynomial in P_3):
  - $T(x) = x^2 \neq 0 \implies x \notin \ker(T)$.
  - $T(0) = x \cdot 0 = 0 \implies \mathbf{0} \in \ker(\mathbf{T})$.
  - $T(1 - x) = x - x^2 \neq 0 \implies (1 - x) \notin \ker(T)$. Only 0 is in $\ker(T)$.
- **(b) NO, T is NOT onto.** Two clean justifications:
  - *Direct counterexample:* Every output $T(a_0 + a_1x + a_2x^2) = a_0x + a_1x^2 + a_2x^3$ has constant term 0. Hence any polynomial in P_3 with nonzero constant term, e.g., $q(x) = 1 \in P_3$, has no preimage under T.
  - *Dimension argument:* $\dim(P_2) = 3 < 4 = \dim(P_3)$, so $\dim(R(T)) \leq 3 < \dim(P_3)$, making it impossible for $R(T)$ to equal P_3.
- **(c)** For any $p(x) = a_0 + a_1x + a_2x^2$, $T(p(x)) = a_0x + a_1x^2 + a_2x^3 = 0 \iff a_0 = a_1 = a_2 = 0 \iff p(x) = 0$. Hence $\ker(T) = \{0\}$ (the trivial zero subspace), so:

$$\dim(\ker(\mathbf{T})) = \mathbf{0}, \qquad \text{and its basis is the } \textbf{empty set } \emptyset.$$

  (Note: By Rank–Nullity for linear maps, $\dim(\ker(T)) + \dim(R(T)) = 0 + 3 = 3 = \dim(P_2)$.*)*

Appendix: Files Scanned & Excluded Topics

All **25 unique files** (out of 29 total, accounting for 4 exact MD5 duplicates) in `past assessments/` were examined:

- **Exams & Solutions:** finalterm_solution.pdf (Sp25 FE), LA+FE+Fall+22.pdf, LA+FE.pdf (Sp22 FE), LA+FE+Spring+24.pdf, LA+Mid+Fall+22.pdf, LA Mid Spring 23.pdf, LA Mid Spring 24 (Final).pdf (+ dup LA+Mid+Spring+24+%28Final%29.pdf), LA Mid Spring 26 (Final) (1).pdf, LA+Midterm.pdf (Sp22 Mid), LA+Spring+23+FE.pdf, MID+A+Solution.docx, MID+B+Solution.docx.
- **Quizzes (all 13 unique):** quiz1 (+ dup), quiz2, quiz3, quiz4, quiz5, quiz6, quiz7 (+ dup), quiz8, quiz9, quiz10, quiz11 (+ dup), quiz13, quiz14.

Excluded as Outside Midterm II Scope:

- **Pre-Midterm I (Chapters 1–2):** Solving linear systems / parameter conditions / LU decomposition / elementary matrix calculations / matrix algebra ($AX = XB$, $AB = BA$, A^{r+s}) / determinant identities & cofactor rules / Cramer's rule proof (D_k) / skew-symmetric diagonal $= 0$ (quiz1–quiz5, Mid Part A sections, MID+A+Solution.docx, LA Mid Spring 26 Part B Q1–Q2 & Q4). *(Master Theorem / consistency bridges are kept in Q5.3.)*
- **Post-Midterm II / Final-Only (Chapters 6.3 & 7):** Gram–Schmidt process & orthonormal bases (LA+FE Q3, Q4(c), LA+Spring+24 Q7, LA+Spring+23 Q10, finalterm_solution P6, P7(a), P10(a), quiz11 P3, P4(b), P7, P8), eigenvalues, eigenvectors, nilpotent eigenvalues & diagonalization (LA+FE Q1, Q2, Q6; LA+FE+Fall+22 Q5, Q9; LA+Spring+23+FE Q6(a),(b), Q7, Q10; LA+FE+Spring+24 Q8(a), Q10; finalterm_solution P8, P10(b); quiz13, quiz14).